Electronic Batch Manufacturing Record (eBMR) Build Outline

Tablets edition and Injectables edition
India (Revised Schedule M) · US FDA · EU GMP

241
controls implemented and tested

8
build phases, each ending in a working demo

14
decisions for you, each with a recommendation

Date	27 September 2026
Status	Revised after two independent critiques (103 issues resolved)
Repository	github.com/maglesT/ebmr · docs/outline/build-outline.md

Plan only: no application code has been written yet. Regulatory content is paraphrased; clause references point to the official texts.

Contents

1. 1. Summary
2. 2. Product scope per edition
3. 3. Regulatory approach
4. 4. Architecture
5. 5. Core logic
6. 6. MBR designer and building blocks
7. 7. Users, segregation of duties and key screens
8. 8. Helpful extras (ranked) and what is deferred
9. 9. Quality and validation strategy
10. 10. Modifiability
11. 11. Phased build plan
12. 12. Risks and mitigations
13. 13. Decisions needed from you

Version 0.2, for your approval. 27 Sep 2026. Plan only: no application code has been written. This version resolves the regulatory review (R-01..R-35) and the logic/UX/buildability review (L-01..L-68); `resolution-log.md` records how each item was handled.

How to read this. Each technical choice is first explained in plain language (*In plain words*), then stated precisely. DI-xx, PR-xx, ST-xx and IN-xx are the 241 controls in review pack v0.9 that the external expert is checking. K-xx/TB-xx and KB-xx/IB-xx are building blocks from the Tablets and Injectables process specifications; G-xx is the canonical generic list (6.6). INV-xx are invariants: rules that must be true at every moment, enforced by the application and again by the database (5.3). S-numbers are rows of the segregation-of-duties register (7.2). B-xx are rows of the India market-resolution table. Status marks on sources: S = secondary source only, U = unconfirmed, P~ = clause number inferred; anything so marked must be confirmed before validation documents cite it.

---

1. Summary

What gets built. Two separate web products that replace paper batch manufacturing records for Indian drug manufacturers, many of them exporters:

- **Tablets edition:** oral tablets from dispensing to finished packs (direct compression, wet and dry granulation, film/functional/sugar coating, bilayer, printing, inspection, blister/strip/bottle packaging, cartoning, serialisation).
- **Injectables edition:** sterile injectables (aseptic vials, ampoules and prefilled syringes, lyophilised vials, terminally sterilised small- and large-volume products, aseptic suspensions/emulsions, BFS/FFS; isolator, RABS, conventional and single-use lines) plus the media-fill (APS) record type.

Each edition is sold, installed, validated and upgraded on its own, with its own server application, web application, database, screens and MBR designer. The designer's palette holds only that edition's blocks: a customer can shape any product within the dosage form but cannot turn one edition into the other.

What is shared. A **compliance kernel** used by both: identity and access (built on natural persons, not just accounts), electronic signatures, audit trail and trusted time, plus the compliance plumbing around them (hashing and sealing, the control register, the market-profile engine, retention, true copies). It is published as versioned packages and pinned exactly by each edition, so the products can never drift apart on data integrity. A site that runs both editions shares one **site-services** store for people, qualifications, material lots and batch numbers (Decision 13). Sharing **recipe-core** (flow engine, expression language, limit model) is recommended (Decision 1).

Release model. Every batch is made under an Indian licence, so the Indian authorised person (AP) releases each finished lot before it is sold or supplied anywhere. That **manufacturer release** gates every market; the US quality-unit disposition and the EU QP certification follow it per market. An AP-released lot may ship to the named EU importer in quarantine while its EU status waits for the QP's certification after importation. A released lot is **suspended** automatically when new evidence links to it.

Market-profile engine. India's Revised Schedule M always applies; each batch adds US and/or EU. For every atomic rule the engine resolves the strictest value across India and the batch's markets, stores which market required it, and applies it at MBR approval, issue, execution, release, sample retention and archiving (IN-52). Rules that cannot be ranked (compendial schemes, label content) use explicit "all apply" and "per destination" methods. Line, room, equipment, person and material-lot rules take the markets of everything that depends on them. A market added after a batch starts is checked against what was actually recorded.

Stack. *In plain words:* one language everywhere, one mainstream database, well-known open-source tools. TypeScript end to end (same checks in browser and server; the server decides); Node.js LTS, Fastify, Zod; PostgreSQL 16+ as the only system of record (Kysely, reviewed migrations, privilege separation, triggers and constraints as a second line); React + Vite, React Flow, XState, decimal.js with UCUM units, pg-boss; Vitest, fast-check, Playwright, StrykerJS; pnpm + Turborepo in github.com/maglesT/ebmr with shared packages published to a private registry.

Phases at a glance (each ends in a tagged release with a demo and passing tests; details in 11):

Phase	Delivers	Demo
0	Repo, CI, control register with target phases, package publishing, first ADRs	CI green; 241 controls listed with target phase and test status
1	Compliance kernel, security architecture, market-profile engine, qualification register stand-in	Person-based four-eyes grant; shared-kiosk entry refused without a fresh badge tap; sign, correct, see the signature invalidated; restore drill with signed restore event; rules resolved for {IN} and {IN, US, EU} on stub lines
2	Recipe engine, expression language, document-style MBR designer, generic step screen, sandbox database, site-gateway core with scale simulator	Build an MBR with a split granulation, fix validation errors, simulate, test-run, approve for India only while US approval is pending
3	Tablets edition end to end with LIMS-lite and deviation-lite; sample BMR as first golden batch	Wet-granulation, film-coated batch with LOD loop, reprocessing addition, left/right IPC with a failing check that holds containers
4	Review, manufacturer release and market dispositions, EU QP package, post-release	AP release then US disposition; EU lot shipped to the importer in quarantine; late deviation suspends a released lot; US added after

Phase	Delivers	Demo
	suspension, India fields, PQR extract, true copies	execution is refused on recorded evidence
5	Injectables edition end to end	Isolator vial batch with EM, interventions (one unlisted), filter tests, APS gate; an action result reported on day 5 holds the batches of days 1-5
6	Remaining integrations with fault injection, dashboards, genealogy explorer, languages, extras	Mock recall; Hindi screens; ERP/QMS/LMS/CMMS adapters exchanging data under injected faults
7	Generated validation pack; load, soak, penetration testing and hardening verification	Validation pack; IQ/OQ run with evidence; load, restore and pen-test reports

Main changes from v0.1: India AP release made a precondition of every market and suspension/recall paths added (5.2); EU export route unblocked (4.6); market additions after execution checked against recorded data, and line-level tightening gated (3.3); correction, void and invalidation separated so a failing result cannot disappear (5.2); shared-device entries bound to a fresh authentication and segregation evaluated per natural person (5.5, 7.2); batch audit-trail review widened to the records the batch relied on (5.4); per-market MBR effectivity (5.2); full APS life cycle and EM linkage from sampling to reporting (5.2); security, restore and archival designed in Phase 1 (4.7, 5.4); phases re-ordered to their dependencies (11).

Honest size: comparable to a commercial manufacturing-execution product; here we build and demonstrate a heavily tested reference implementation with simulated equipment and systems (11.1).

2. Product scope per edition

2.1 What customers configure (both editions)

In plain words: customers assemble processes from validated blocks and enter their own values; they cannot write code or weaken a regulatory rule.

- **MBRs** from typed blocks: stages, operations, steps, formula/BOM, equipment-class requirements, parameters with tiered limits (registered / internal action / alert, 5.2.4), calculations, IPC plans, timers and hold times (each with a validation reference), sampling, signature points (each with a meaning such as PERFORMED, VERIFIED or WITNESSED), QA hold points, branches, bounded loops and replicate-per-sub-lot sections. **Library elements** and **unit-procedure templates** (tablets UP-10..UP-90; injectables A.3): reusable compositions verified once.
- **Generic masters + variant parameter sets** (strength, tablet weight, fill volume, artwork per market). Approval covers the pairing; only the discrete batch sizes listed in the master (PR-01, IN-13).
- **Master data** (materials, suppliers, equipment and instruments, rooms/lines, tooling, templates, reason codes, units, custom fields; a custom field used in logic is GMP data), **organisation** (sites, roles and bundles, approval routes that can be extended but never reduced, deviation-class competence, training gates, delegations, languages), **site values** in rule tables (tighten-only, ST-01), and the **policy switches** the expert and customer may change (3.5).
- **Never changeable** (features spec E.2 plus the India floor): the audit trail; the minimum signature set; the locked separations (7.2), including the approved-technical-staff gate (S17) and reviewer independence (S10); commit-before-evaluate; no pre-filled measured values; no labelling-reconciliation waiver for Indian-made product; the training gate at every action; equipment and **area** logbooks; approved specimens for every printed component in a packaging master (IN-14); the processing line-clearance checklist (IN-21); the India AP manufacturer release before any market's release (IN-05).

2.2 Tablets edition: variants in v1

Code	Variant	Key controls in the process
DC	Direct compression	Blend-uniformity plan
WG-HS / WG-FB	Wet granulation (high-shear + dryer; fluid-bed)	Binder use-by, endpoint, LOD loop, filter bags (IN-36)
DG-RC	Roller compaction (slugging legacy)	Ribbon checks, fines-recycle limit
UNC	Uncoated and sub-types (chewable, dispersible, orodispersible, soluble, effervescent)	Sub-type selects disintegration rule set
FC / FC-F / SC	Film, functional and sugar coating	Weight-gain endpoint; US controlled release needs a release-rate result per active (PR-54)
BL	Bilayer / multilayer	Layer weights kept apart from whole-tablet statistics
PK-BL, ST, BT, SEC, TER, SER	Blister, strip, bottle, secondary, tertiary, serialisation (US, EU, India Schedule H2)	One packaging record per packaging order; market-specific orders from one bulk batch, with a full clearance between orders (5.2)
Modes	Potent/hormonal/cytotoxic, hygroscopic/effervescent, light-sensitive; **controlled substance** (Decision 14)	Containment, RH limits, exposure timers; dual custody and zero-difference reconciliation of the controlled API

Sub-lots split, merge and replicate with genealogy (K-25, 6.2).

2.3 Injectables edition: variants

Code	Variant	Proposed v1 depth (Decision 7)
V-1	Aseptic solution in vials	Full, golden batch
V-2	Aseptic glass ampoules (100 % CCI)	Full

Code	Variant	Proposed v1 depth (Decision 7)
V-3	Prefilled syringes/cartridges from RTU tubs	Blocks in v1; golden batch v1.1 (combination-product detail unverified)
V-4	Aseptic fill + lyophilisation	Full, golden batch
V-5	Terminally sterilised small volume	Full, golden batch
V-6	Terminally sterilised large volume	Blocks in v1; golden batch v1.1
V-7	Aseptic suspensions/emulsions	Blocks in v1; golden batch v1.1
V-8	BFS/FFS aseptic	v1.1
V-9	Aseptic + post-aseptic heat treatment	Full (adjunct terminal mode)
V-10	APS (media fill) record	Full, golden batch, own life cycle (5.2)

Technology profiles (ST-74): open/closed isolator, open/closed RABS, conventional A-in-B (needs a contamination-control-strategy justification), BFS, FFS, closed/single-use.

2.4 Out of scope for v1

Continuous manufacturing (ICH Q13); capsule filling (Decision 7); biologics (EU Annex 2), investigational products, API manufacture, liquids, semi-solids; sterile powder filling, cell therapy, radiopharmaceutical specifics; the parametric-release path until EU Annex 17, the FDA 2010 guidance and the compliance policy guide are verified (ST-61); WHO-GMP, UK and other markets (a product-master "PIC/S-member destination" flag is treated as EU for line-level sterile rules until they are added, 3.3); the enterprise hub and cross-site MBR distribution (topology option C, v1.1; 4.2); full ERP/LIMS/QMS/LMS/CMMS/EMS functionality (integrated, with minimal stand-ins, Decision 11); device-side offline signing (Decision 12); AI features and a graph analytics copy; migration of legacy records (DI-48 kept for later).

2.5 Assumptions (please correct any that are wrong)

1. Every batch is made in India under an Indian licence; India is always in the market set and cannot be removed (IN-52). The Indian AP release is required before sale or supply anywhere, exports included (Schedule M Part I 11.3.8).
2. Export markets in v1 are US and EU. EU batches are normally certified by the EU importer's QP after importation, outside this system; the Indian site performs the manufacturer release, supplies a QP batch package, may ship the lot to the named importer as quarantined goods, and records the certification reference received back. In-system QP certification is a tenant switch, OFF by default (DI-39, PR-55).
3. v1 topology: one installation per **site** per edition and environment (development, validation, production), run on site or in a single-tenant private cloud. A company with several sites runs several installations; lots move between them by signed transfer packages (4.2). A site running both editions shares one site-services store (Decision 13). Contract manufacturers can separate client data inside an installation (4.4).
4. ERP owns the item master, orders and inventory valuation; WMS owns quantities on hand; the eBMR owns MBRs, execution, batch records, genealogy and material **usage status** (IN-24(a)); LIMS supplies results; LMS and CMMS supply qualification and calibration data where integrated. Section 4.6 names the system of record per data item.
5. Customers qualify their own installation (IQ/OQ/PQ) using the pack we generate.
6. English is the master language; Hindi and regional languages are display languages whose MBR translations are approved with each version. Free-text GMP entries on batches for the US or EU are in English or carry a verified translation (7.4).
7. Pharmacopoeial values (USP, Ph. Eur., IP 2026) ship as master data marked unverified until customer QA confirms them.
8. The user's sample BMR is a coverage test and the first golden batch, not a template.
9. Export-only products not approved in India need an Indian export permission recorded per product, strength and destination; the exact instrument and label particulars (Rule 94) are status U until counsel confirms (India G-9, G-

10).

3. Regulatory approach

3.1 The control register is data

In plain words: the 241 rules under expert review are stored as structured files in the repository; code, tests and validation documents are generated from the same list, so they cannot silently disagree.

- `compliance/controls/` : one YAML file per control (DI-01..50, PR-01..60, ST-01..79, IN-01..52, plus draft IN-53 if Decision 14 = A). `compliance/requirements/` : the 678 clause-level rows (n-numbers) with verification status (checked, secondary, unverified, draft).
- Each control file holds: ID, title, editions, type (ENFORCED / CONFIGURABLE / DESIGN_CHOICE), markets, source clauses with verification markers, requirement rows, paraphrased behaviour, **implemented_by** (modules, blocks, guards, invariants, database constraints, rule keys, policy switches), **tests**, **target_phase** (the phase whose exit must prove it), **expert_decision** (PENDING / FOLLOW / DONT_FOLLOW / MODIFY, reviewer, date, pack version, comment) and build status (SPECIFIED / IMPLEMENTED / VERIFIED). Where this outline proposes to go beyond a catalog control (e.g., blocking US issue without the 11.100(c) record, DI-15), the file carries a `proposed_modify` note so the expert decides it.

```
yaml id: DI-25 title: Audit-trail review within batch review editions: [tablets, injectables] type: [ENFORCED] target_phase: 4 sources: [{ref: "21 CFR 211.192", status: checked}, {ref: "Annex 11 draft s.12", status: draft}] implemented_by: {modules: [M02, M19], blocks: [G-25], invariants: [INV-35, INV-48], policies: [QA_INPROCESS_COUNTS_AS_INVOLVED]} tests: [T-DI-25-01, T-DI-25-02, T-DI-25-03] proposed_modify: "Scope widened to audit events on linked records (R-17/L-10)" expert_decision: {status: PENDING, pack_version: "0.9"}
```

3.2 Trace chain

Clause (n-row) → control → user requirement (URS, generated per edition) → design element (module, block, guard, invariant, constraint, rule key) → automated test named with the control ID, e.g. `[DI-25][INV-35] a person who signed a QA hold point cannot sign the audit-trail review` → result and evidence. CI rebuilds the traceability matrix on every change. It fails if a FOLLOW, MODIFY or pending ENFORCED control **whose target phase is at or before the current phase** lacks a passing test in its editions, if a test cites an unknown control, or if a DONT_FOLLOW control still has active enforcement. Every catalog "Test" line becomes at least one automated test (at least 241 control-level tests).

3.3 How the market-profile engine works

In plain words: all market differences live in one versioned set of rule rows; one small engine picks the value that satisfies every market and records why, instead of "if US… if EU…" scattered through the code. Where "strictest" has no meaning, the engine says so and uses a stated method instead.

3.3.1 Rule model

- **Keys are atomic:** one dimension and one unit each. A composite regulatory rule becomes several keys, each with its own comparison method. Examples: reserve samples = quantity basis, retention period and yearly visual check (B-02, B-26); Grade A particles = 0.5 µm limit, 5 µm limit, monitoring mode, sample flow; filters = maximum working time, single-batch use, validated lot count (B-17, ST-53); EM viable limits evaluated per sample (IN-42 design choice) with the per-grade values as separate keys.
- **Rows, not columns:** each value is a row (key, market, optional predicate, value, unit, **floor type** of that value, source and verification status, effective-from, effective-to). The floor type sits on the value, because the same key can be hard in one market and guidance in another: HARD (binding number), GUIDANCE (e.g., Annex 1's 10 Pa step), EXAMPLE (e.g., FDA's "e.g., at least 12.5 Pa"), DESIGN (our product choice). A new market is new rows, not a schema change (L-58).
- **Predicates** select a value inside one market from product or site attributes, and the branch used is stored: licence form (Form 25 vs Form 28, retention, U), OTC product exempt from expiry dating (US retention), manual aseptic process (per-operator APS, ST-30), Schedule H2 category and pack date, PIC/S-member destination.

- **Comparison methods:** SMALLER_STRICTER; LARGER_STRICTER; ORDERED (categorical list); REQUIRED_IF_ANY (booleans); **SET_UNION** (the result is the union: sterility pull points, time-limit lists, record content items, ST-66); **ALL_APPLY** (compendial acceptance schemes that cannot be ranked, e.g., uniformity of dosage units, weight variation, disintegration, dissolution: every applicable criterion is evaluated on its own sampling plan with its own rounding rule, the sampling plan is the union of what each criterion needs, the result passes only if every criterion passes, and each criterion's result is stored with its governing market); **PER_DESTINATION** (not orderable, e.g., label content: the author chooses the governing source for each pack destination and approval is blocked until a regulatory decision reference is attached). The validator refuses a scalar method on a multi-component key.
- **Tables:** kernel keys (retention, signatures, time) and per-edition keys (`tablets.*`, `sterile.*`) are separate tables, each versioned on its own, so a sterile change never creates a change-control impact for Tablets customers (L-57). Tables reach on-premise installations as signed data packages, separate from platform releases.

3.3.2 Resolution, scopes and what is stored

- **Resolution:** for each atomic key, the value that satisfies India plus every market in scope, using the key's method. INV-28: for every atomic key, the resolved value is at least as strict as India's.
- **Scopes:** PRODUCT and PACK (the batch's markets; labels per pack destination); SITE; **LINE, ROOM, EQUIPMENT, PERSON** (the union of markets over every product qualified on that line, room or equipment, or, for a person, on the lines and rooms they are qualified for; a PIC/S-member-destination flag counts as EU for line-level sterile rules); **MATERIAL_LOT** (resolved at each consumption against the consuming batch's market set, because one lot feeds batches with different markets). Lines, rooms, equipment and qualifications are edition entities, so the kernel engine asks a `ResolutionContextProvider` port that each edition implements; the kernel imports nothing from the editions (L-59).
- **Consumption guard, added item:** the lot's recorded release route (identity-tested, or India-only vendor-history route under 14.15) must satisfy the consuming batch's resolved identity-test rule (IN-24(c), B-09). A vendor-history lot dispensed into an {IN, US} batch is blocked.
- **Stored at every resolution point** (MBR approval, batch issue, each execution evaluation, release, sample retention, archiving): resolved value, governing market(s), predicate branch, rule-table version and inputs. Every limit or timer evaluation stamps the limit version (ST-01).
- **Site values** may only be stricter. A value looser than a HARD number is refused; loosening a GUIDANCE value needs a contamination-control-strategy reference and QA approval and is shown on every affected batch (ST-01); an EXAMPLE value adopted as our default is loosened the same way.
- **Source corrections are not loosening:** a shipped pharmacopoeia or rule value that turns out to be wrong is corrected as a source correction (citation, two QA signatures, vendor notified, new data version). Tighten-only applies to site values, not to source corrections (L-38).

3.3.3 Changing a batch's market set

In plain words: re-computing rules cannot change what already happened on the floor, so adding a market later is checked against the record, and removing one is never a way around a failure.

- **Before issue:** free; the rules are simply resolved again.
- **Adding a market after issue** needs a QA e-signature and a change record and runs a **market-add evaluation**, which becomes a hard release-gate item for that market, not a QA judgement. It re-evaluates, against the recorded data: the MBR version's validity for the new market (5.2 MBR); every key that became stricter, including verifier counts per step (one-verifier mode is acceptable for the US only on 211.68(c) step types), IPC intervals, hold times, filter use and PUPSIT records, the identity-test route of every consumed lot, line APS and EM status under the new acceptance rules, EM limits, label rules, reserve-sample quantity and retention. Each item ends MET, UNMET_RECOVERABLE (still achievable, e.g., an extra test, more reserve samples, a longer retention) or UNMET_FINAL (cannot be achieved any more, e.g., a sampling interval, an identity test that had to precede use, a filter used across two batches). Any UNMET_FINAL item makes that market's disposition **INELIGIBLE**, a terminal state; UNMET_RECOVERABLE items stay open release-gate items until met.

- **Removing a market after issue** needs a linked deviation and a QA e-signature. The original resolution stays in the record and nothing already evaluated is re-labelled; the removal is shown in the release cockpit and the PQR extract, and a removal after a failure against that market's rule is flagged "removed after failure".
- **Packaging orders** can be created only for destinations inside the batch's market set; otherwise the market-add evaluation runs first.

3.3.4 When a line, room, equipment or person rule tightens

- Qualifying a product for a stricter market on a line, room or piece of equipment (the first EU product on a line, for example) is a **gated event**. Before the first batch for that market can be issued, the engine re-evaluates the evidence inside the current qualification window against the new rule: APS runs back to the most recent initial or repeat validation (a run once recorded as "one positive in 12,000, investigate" is a failure under the EU rule), shift coverage, EM results and trends (e.g., FDA ISO 8 100/50 versus Grade D 200/100), requalification and HEPA dates, operator APS participation, filter reuse counts. Any failing item leaves the line NOT_QUALIFIED for that market until requalified (e.g., three consecutive successful APS).
- **Loosening** (a product or market leaves a line) happens only through change control, applies only to runs and results after its effective date and never turns a past failure into a pass.
- Decision 2 decides whether tiered FDA/India APS acceptance is offered at all; if it is not (recommended), APS acceptance is "any contaminated unit fails" on every line and the tightening case above cannot arise for APS.

3.3.5 Rule-table updates reaching approved MBRs

- Rule rows carry **effective-from dates** equal to the legal effective date of the source (including any transition the source itself grants).
- At batch issue the MBR content-minima validator re-runs against the table version in force. An MBR limit looser than the resolved value blocks issue. There is no QA override to issue with a limit looser than a rule already in force; transitions exist only where the source itself grants them.
- A new table version produces an impact list of MBRs and site values, turned into tracked change tasks due before the rule's effective date (e.g., every MBR citing IP 2022 when IP 2026 applies, IN-33).
- Running batches keep their issue-time table version; applying a newer one mid-batch needs a QA change-control decision and an audited re-resolution (5.8).

3.3.6 Behaviours as keys and worked examples

- **Behaviour keys** are boolean or categorical keys read by guards and the MBR validator. Corrected in v0.2 so India's column is never skipped: **area logbooks are always on and ROOM-scoped** (Schedule M requires chronological records of major equipment and of the areas where product was processed, 17.3.10.17 P~; IN-25(a); an area logbook active only during EU batches would leave gaps); **approved specimens for every printed component are required in every profile** (IN-14; PR D-1's "US in set" resolution is superseded because India always applies; N/A only by a recorded MODIFY on IN-14); **the processing clearance checklist is locked on** (IN-21; PR D-5's "configurable, default on" is superseded); reconciliation waiver never offered (IN-17). A CI check fails if any behaviour key's IN value has not been reviewed against the India catalog.
- **Retention** (kernel keys; the latest of all that apply, DI-47, PR-59, IN-08): Schedule M expiry + 1 y (always); Drugs Rules conditions of licence, predicate on licence form: Form 25 (Rule 74) 5 y from manufacture for Schedule U records, Form 28 (Rule 78) test records expiry + 2 y where expiry is labelled, else 5 y from manufacture (U); US expiry + 1 y, or 3 y after distribution for OTC products exempt from expiry dating; EU the later of expiry + 1 y and QP certification + 5 y, kept **open** until the certification date is received; EU batches never certified: the later of expiry + 1 y and disposition + 5 y, pending QA confirmation (PR F-2); aborted or cancelled batch records: the later of the nominal expiry the batch would have had + 1 y and the abort/cancel date + 5 y (design default, QA to confirm); DSCSA serial and transaction reconciliation records for US batches: at least 6 y (s); EU validation raw data supporting the authorisation: while it is in force and at least as long as every batch released on it (4.12). **Retention dependency:** a shared or supporting record (APS runs, EM sessions, equipment and area logbooks, component, container, closure and labelling lot records, MBR versions, configuration, training evidence, signature register) inherits the latest retain-until of every record that depends on it, recomputed whenever a dependant's expiry or certification date changes (211.180(b); DI-47; INV-22 test: a component lot used by two

batches with different expiries follows the later one). Audit trails and signatures follow their record. Legal and DI-investigation holds override disposal.

- **Other examples** (IN-52 test, table B): excipient identity test on every lot for US or EU batches and by default for India-only batches (B-09, IN-24(c)); compression sampling against the in-process specification at most every 30 min, and station-wise checks at fixed MBR intervals on multi-point presses (B-03); IP weight bands (B-14, U) inside the ALL_APPLY scheme with the release UDU test; APS "any contaminated unit fails" per Decision 2 (B-10); Grade B in-operation 5 µm limit 2,900/m3 from India (B-13); Grade A/B requalification including the filter leak test at 6 months or less in **every** profile, a product choice stricter than India-only B-16 (block at 12, warn at 6), and C/D at 12 months or less (ST-14, Annex 1 4.32); pre-use filter test always, after sterilisation when EU is in the set (B-17); uncoded manufacturing date on India-domestic packs (B-15), treated as **U for export packs** until Rule 94 question G-9 is answered, with a mandatory "India export particulars" checklist next to the destination's approved artwork.

3.4 How the expert's Follow / Don't follow / Modify decisions flow into the build

1. **Capture:** decisions from the PDF are imported into each control's `expert_decision` as a reviewed commit (the decision history is auditable).
2. **FOLLOW:** build as specified. **MODIFY:** control text changed to the agreed wording (diff reviewed), implemented_by and tests updated, new control version.
3. **DON'T FOLLOW:** for a design choice, the policy switch is set off in the vendor profile or the feature is not built. For a control resting on binding text, nothing is dropped silently: the build shows which markets would no longer be met and asks you to confirm; if confirmed, it is disabled only for the affected profile and the release notes say so.
4. Architecture-level changes get an ADR.
5. **Timing:** Phases 0-2 can start before the pack returns, because design choices are switches defaulting to the pack's proposal and the CI gate only covers controls whose target phase has been reached; controls still pending at a phase end are listed in its release notes.
6. The review-pack builder reads the register, so a v1.0 pack with decisions (including this outline's `proposed_modify` items) is regenerated automatically.

3.5 Design choices are switchable, versioned and pinned

In plain words: the few settings customers may change are versioned like documents, and a looser setting never reaches a batch that was already issued.

- `compliance/policies/` lists each switch: key, controls, default, allowed positions, **level** (vendor profile set from the expert's decisions, or tenant setting owned by customer QA under change control), what loosening requires, and tests for every position.
- **Versioning and pinning** (L-25): each switch is a versioned configuration item with author, approver, audit trail and effective date (DI-09). A batch pins the policy versions in force at issue. A **loosening** applies only to batches issued after its effective date; a **tightening** may apply to running batches at once. Every disposition stores the policy versions it evaluated. The RBE rule-set version is fixed at production approval.
- **Switches:** checker may also approve an MBR (S4, default no); one human verifier for validated automated operations on 211.68(c) step types (S5a, default **two persons on every batch**; a batch run in one-verifier mode can later add US only if every such step is a 211.68(c) type); second person on packaging line clearance (S24, default on, always on for blister/strip); approval level for critical-data corrections (S25); QA approval of each void (S28, default on); destruction witness (default on); charge-in verifier within a time window from a room terminal (PR-21 design choice, default off); relaxation of signature components within one continuous session of the session owner (DI-38, default off; never for disposition, certification or MBR approval; details 5.5); whether QA in-process checks (IPC or AQL sampling) make a QA person "involved" in a batch (default yes, 7.2); review by exception (default full review until the MBR meets its confidence criterion, Decision 10); release with an open investigation by e-signed justification (PR-53, default blocked; excluded cases in 5.2.7); CAPA blocking release (PR-51, default no); in-system QP (default off); English-or-verified-translation for free text on US/EU batches

(default on, 7.4); DSC/eSign per signature meaning (default off until a licensed provider is configured, Decision 3).

- **Not switches any more:** reviewer and disposition-signer independence (S10) and the approved-technical-staff gate (S17) are locked (7.2); the training gate is a hard block on every batch, because every batch is India-licensed (IN-04(e)).

3.6 Absorbing regulatory change

- **Watch list** (`compliance/watch-list.yaml`, owner and next check): final EU Annex 11, Chapter 4 and Annex 22; EU Annex 16 text (not in the inputs; QP package content to verify); IP 2026 amendments; Schedule H2 data after G.S.R. 506(E); Schedule M corrigenda and the unreadable pages (IN F-1 to F-12); CSA status; DSCSA and EU safety-feature details; **NDPS Act 1985 and NDPS Rules** (manufacturing-record and stock-account duties, for Decision 14); **Digital Personal Data Protection Act 2023 and its Rules** (personal data held for GMP retention); **CERT-In Directions of April 2022** (incident reporting, log retention in India, clock synchronisation); Indian export permission practice and Rule 94 particulars (G-9, G-10). The last four are not in the inputs: verify before citing.
- **Workflow:** update requirement rows → tool lists affected controls (for Annex 11 and Chapter 4 the DI catalog already names them: DI-02, DI-08, DI-11, DI-13 to DI-19, DI-22, DI-23, DI-25, DI-36 to DI-39, DI-41, DI-43, DI-45 to DI-47, DI-49) → controls modified with expert review → rule-table or code change → release with a validation delta.
- **Change classes** (E.7): platform release (full regression + customer acceptance); rule-table version (kernel or edition table; impact list and change tasks, 3.3.5); configuration item or policy version; MBR or library element.
- **Drafts:** stricter draft expectations that conflict with no binding text are already followed. Three are deliberately not: removing the stand-alone QP e-signature clause, audit-trail review after release, MFA at every signature as default (DI catalog section b). These are re-checked first when final texts appear.
- **In-flight batches** keep their issue-time rule-table and policy versions unless QA re-resolves through change control (5.8).

---

4. Architecture

4.1 Web application

In plain words: people use a browser; the server makes every decision; the database double-checks the important rules.

Browsers on shop-floor panel PCs and cleanroom-rated tablets (kiosk mode, badge reader, 2D scanner) and on office PCs, each a **registered device** with a class: PERSONAL, SHARED (kiosks, panel PCs, cleanroom tablets) or GRADE_A_B (shared, grade-rated). Stateless Fastify API processes plus workers (timers and sweepers, outbox relay, read models, audit anchoring, integrity and seal verifiers, PDF rendering, contingency-pack generation). Every command is validated and executed on the server in one database transaction; the UI never decides whether a step may start, a value is in limits or a signature is valid.

4.2 Topology, deployment and continuity

Option	Suits	Notes
A. On-premise at the site (default)	Most manufacturers	App + PostgreSQL with synchronous standby on site; customer controls upgrades; survives WAN loss
B. Customer's private cloud, one tenancy per site	Groups with a cloud landing zone	Single-tenant, Indian region; the site depends on its WAN link, so the paper-fallback kit must be pre-staged (below)
C. Site cells + enterprise hub	Large groups	**v1.1.** Hub holds global templates and master data sent to sites as signed packages; sites adopt them locally

- **v1 topology** (L-41): one installation per site per edition and environment. A company with several sites runs several installations. **Inter-site transfer** of intermediates or bulk: the shipping site produces a signed lot-transfer package (status and disposition or CoA, genealogy excerpt, remaining hold time, rule resolutions that travel with the material); the receiving site creates a local lot in QUARANTINE, and local QA accepts it before use. Cross-site genealogy queries are federated and carry an as-of point. In v1.1 a hub-approved MBR becomes usable at a site only after site QA **adopts** it with a local signature.
- **Site services** (Decision 13, recommended option A): a site running both editions has one site-services database holding persons, accounts, credentials and badges, qualifications and training, material lots with their usage status, the batch-number allocation register and the device registry. Each edition keeps freshness-bounded mirrors for its guards and consumes material through short-lived **reservations** granted by site services, so every edition transaction stays local and atomic (INV-13, INV-14). A lot status change in site services revokes open reservations and is pushed to the editions; consumption against a revoked or expired reservation is refused, and a revocation that arrives after consumption raises an exception. An excipient lot rejected for one edition is thus unusable in the other, and batch numbers are unique across both (IN-18). A single-edition site runs the same site-services database on the same server.
- **Network loss.** The site server keeps the floor running through WAN or cloud outages. If a terminal loses its link, the unsent entry stays on screen with a blocking "not saved" banner (it is not yet a record) and is re-sent with the same idempotency key. A pending entry stays bound to the person who authenticated it and to that authentication event; if the terminal changes user it is discarded, and a local notice lists it for the incident record (L-67). The gateway buffers instrument data with source timestamps. **Terminals never sign offline** (Decision 12).
- **Paper fallback that works when the site server is down** (L-48, DI-43). At issue and at every stage start the server pre-generates a numbered contingency pack (PDF with hash) per active batch on a local print server or edge cache; form numbers are allocated in the register and reconciled before batch closure. Aseptic filling has an immediate, pre-staged fallback. Back-entry records **three identities**: the performer who signed the paper, the transcriber and the verifier (DI-30); the transcriber's e-signature never replaces the performer's handwritten one. The contingency form is a declared hybrid point in the DI-01 register, and the signed paper stays the original record for those steps unless replaced by a linked, verified true copy (DI-41). Every guard is re-evaluated against the historical state; a failing guard becomes an exception and a deviation, never a silent pass or a block

on back-entry. Values unsaved when the server failed are listed in the incident from the terminals' local notices (DI-29).

- **High availability and restore** (designed in Phase 1, L-47). Quorum synchronous commit to standbys; if synchronous replication is lost the cluster falls back to asynchronous replication with an alarm, and the degraded period and its RPO are recorded. A point-in-time restore writes a **restore event**, anchored externally, that names the discarded anchor range, so the verifier accepts chains that end at the restore point; a lost-window incident is created automatically listing the batches active in the window, with re-entry through the late-entry or paper path; batch-number, form-number and per-batch counters resume above the highest value found in anchors, issued contingency packs and ERP acknowledgements, so no identifier is ever reused. Backups follow DI-45 (retention-grade copy plus operational snapshots, separate and encrypted); restoring needs two-person approval.

4.3 Monorepo layout and package publishing

`ebmr/ compliance/ controls/ (241 + decisions), requirements/ (678), rule-tables/kernel/, rule-tables/tablets/, rule-tables/sterile/, floors/ (ST-01), pharmacopoeia/, policies/, catalogs/ (signature meanings, exception taxonomy, reason codes, units), watch-list.yaml packages/ kernel/ persons and identity, access, e-signature, audit trail, time, hashing/sealing, idempotency, outbox, retention, true copies, control-register loader, market-profile engine; sql/ kernel.* site-services/ persons, qualifications, lots and reservations, batch-number register, device registry; sql/ site.* expr/ expression language recipe-core/ flow model, compiler, validator, engine, simulator, limits, generic blocks, lifecycle machines integration-sdk/ adapter contracts for every external system ui-kit/ design system, signature dialog, audit viewer, review components, i18n validation-kit/ URS/risk generators, trace matrix, PDF test records test-kit/ golden-batch harness, invariant SQL library, bypass tests apps/ tablets-server/ tablets-web/ blocks TB-xx; sql/ tab.* injectables-server/ injectables-web/ blocks IB-xx; sql/ inj.* site-gateway/ scale/printer drivers, OPC UA client, buffering, device identity (edition-agnostic) simulators/ scale, press, coater, autoclave, lyophiliser, particle counter, EMS/BMS, access control, ERP/LIMS/QMS/LMS/CMMS stubs, printers validation/tablets/ validation/injectables/ docs/adr/ docs/architecture/ docs/user/ docs/regulatory/ tools/verifier-cli/ tools/review-pack/ tools/bench/ infra/`

- Lint-enforced boundaries: apps never import each other; packages never import apps; `kernel` imports nothing else in the repo; `recipe-core` may import `kernel` and `expr`.
- **Exact pinning that really works** (L-57): shared packages (kernel, site-services, recipe-core, expr, ui-kit, integration-sdk) are published to a private registry with Changesets; each edition depends on exact published versions, never on the workspace copy. Maintenance branches exist per kernel minor line with a written backport policy. CI builds each edition against its pinned versions plus a canary build against HEAD. Kernel migrations are versioned with the kernel, so a kernel change reaches an edition only through that edition's change control.

4.4 PostgreSQL

- **Databases:** `ebmr_tablets` and `ebmr_injectables` (each with `kernel.*`, the recipe schema and `tab.*` or `inj.*`), `ebmr_site` (site services), and a physically separate `ebmr_sandbox_<edition>` for test runs and training, whose role has no grants on any GMP database; master data is replicated into it read-only (L-52, DI-05).
- **Rows:** every GMP row has a time-ordered UUIDv7 key, tenant and site IDs, a row version for optimistic locking and creation metadata; exact `numeric` values with UCUM unit codes, never floating point; no DELETE outside the retention function (5.4).
- **Tenancy** (L-43): a tenant is the legal owner of records. Site-owned resources (equipment, rooms, logbooks, occupancy, cleaning, clearance) belong to the site tenant and are visible to everyone working there; client-owned content (MBRs, formulas, batch data) is protected by row-level security. Clearance screens and logbooks show a previous-product code and hazard class that do not identify another client, so PR-25, IN-21 and IN-26 checks still work. A person may hold grants in several client tenants; involvement (S10) is computed per batch.
- **Privileges** (L-55): the table owner cannot log in; migrations run only through the pipeline. The application role cannot UPDATE or DELETE audit, signature, data-point or genealogy rows and has **no UPDATE right on status**

columns: state transitions run only through SECURITY DEFINER functions that check the transition table. Only an **authentication-service role** can insert authentication events (single-use, expiring, bound to person, device and signature challenge). Signatures are inserted only by a SECURITY DEFINER function that consumes an unused authentication event of the same person for the same challenge. Break-glass superuser access needs two people and is logged by pgaudit to a sink administrators cannot write (4.7).

- **Volume and partitioning** (L-46): batch-owned tables are range-partitioned on the batch's UUIDv7 month; audit data by record-closure period, so closed periods can be archived as a unit. Sizing target for design: 100 batches a day at one site and a mid-range 20,000 data points per batch give about 7 x 10^8 data points a year, roughly 180 GB a year at 250 bytes per row including indexes, plus a similar volume of audit events once insert-only tables reference row hashes instead of copying rows (5.4). Closed records therefore move to verified archive packages rather than staying online for 15 years. High-frequency instrument data stays in the source system, with per-interval statistics, excursions, cycle summaries and a hash-checked reference in the record.
- **Serial numbers** (L-53): unit-level serials and aggregation stay in the serialisation L3/L4 system; the eBMR keeps pool IDs, counts, the reconciliation and a hash-referenced L3 report.

4.5 Graph question: verdict and benchmark

In plain words: a graph database was considered for batch genealogy and rejected; the normal database answers the same questions fast enough and keeps one validated platform.

PostgreSQL is the only system of record. Genealogy is an insert-only link table (mistakes neutralised by reversal links), indexed both ways, traversed by recursive SQL with set semantics, depth guard and cycle detection behind one `GenealogyService`. MBR flows are in-memory graphs persisted as immutable hashed JSON. A graph database is acceptable later only as a rebuildable, non-GMP analytics copy. Benchmark here (PostgreSQL 16, 4 vCPU; 9.73 M lots, 20.0 M links, hub-heavy): backward trace 0.03 ms average; forward trace from the worst hub excipient lots 48 ms worst; supplier recall over 447,139 lots 0.86 s; audit write cost about 0.6 ms per write at about 6,000 audited commits/s (architecture 1.5). Drug genealogy is shallow (4-8 steps). Recall and impact results are stored and signed by QA with the as-of point, query version and hash.

4.6 Integrations: adapters, site gateway, simulators

In plain words: each outside system connects through a small replaceable adapter; the table below says which system owns which data and how fresh our copy must be before a GMP decision may use it.

`integration-sdk` defines one contract per system type (messages, idempotency keys, error classes, reconciliation report). Inbound data enters only as domain commands through the same checks and audit trail, under integration identities that can never sign (DI-14, DI-31); outbound events use the transactional outbox. The **site gateway** at the automation boundary drives scales and printers, speaks OPC UA to presses, coaters, sterilisers, lyophilisers, particle counters and packaging lines, buffers with source timestamps and stamps device IDs from each device's enrolled identity (DI-20, DI-32).

System of record and freshness (L-44). Inbound copies are **system-of-record mirrors**, updated by commands that carry the source time and a feed heartbeat; they are not read models. Every GMP guard checks a freshness bound per data class and **fails closed** when the mirror is stale for qualifications, lot status, calibration and the QMS status check at release; other stale data is flagged.

Data	System of record	In the eBMR	Stale mirror
Item master	ERP	Mirror	Flag
Material usage status (quarantine, released, rejected, retest)	**eBMR** (IN-24(a)); LIMS supplies results	Authoritative	n/a
Lot quantity on hand	WMS/ERP	Reservations and consumption postings	Flag; consumption uses the reservation
Qualifications and training	LMS if integrated, else site-services register	Mirror or authoritative	Fail closed
Calibration and maintenance due dates	CMMS	Mirror; as-found failures pushed in	Fail closed

Data	System of record	In the eBMR	Stale mirror
Deviations, CAPA, change records	QMS if integrated, else built-in stand-in	Status fetched with a version token before release	Fail closed at release
EM, particle, pressure, T/RH results	EMS/BMS, LIMS	Imported per event	Feed gap = alarm
MBRs, batch records, genealogy, dispositions	eBMR	Authoritative	n/a
Unit serials and aggregation	Serialisation L3/L4	Pool IDs, counts, reconciliation, report hash	Fail closed at release

A mismatch between the ERP order's BOM and the MBR BOM blocks batch issue.

System	Direction	Key behaviour
ERP / WMS (ISA-95, B2MML)	both	Orders and BOM in; goods movements, yields and lot status out. **Shipping rule** (INV-42): no shipment of a lot without a current (not suspended) India manufacturer release; to a US consignee only after US disposition; to the EU **only to the EU importer named in the technical agreement, as quarantined goods, while EU status is AWAITING_QP_CERTIFICATION**; ERP/WMS never sets EU saleable status, which follows the importer's QP certification; overdue certification (default 60 days) raises an escalation task
LIMS	both	Sample requests out; results, specifications and OOS status in; a revised result after review forces re-review and, after release, suspends the release (5.2)
QMS	both	Deviations, CAPA and change records linked both ways; status checked with a version token at release
LMS / CMMS, EMS / BMS	in	Per the system-of-record table; an as-found calibration failure lists every batch since the last good calibration
Access control / airlocks	in	Entries to graded areas reconciled against the aseptic roster; an unreconciled entry is an exception (L-33). Without this integration, entrants badge in at the gowning-room terminal
SCADA / historian (OPC UA)	both	Setpoints read back before start; cycle data, trends, alarms; recipe version checked
Scales, lab instruments	in	Stable reading, raw string, device ID; calibration and daily check at point of use
Printers, coders, vision (GS1)	both	Variable data from the record, job bound to the packaging order, voided if batch data change; challenge results
Serialisation L3/L4 (OPEN-SCS, EPCIS)	both	Serial pools per packaging order, commissioning, aggregation; zero unexplained serials before release; decommissioning task on rejection
EU importer / QP	both	**QP batch package** out; structured certification message in (5.2)

Simulators (`apps/simulators`): scale with stability and noise; press and coater publishing OPC UA tags; autoclave and lyophiliser cycle records; particle counter and EMS/BMS feeds with configurable excursions; access-control events; ERP, LIMS, QMS, LMS and CMMS stubs with consoles; printers rendering to PNG/PDF. All can inject faults (feed gap, stale value, recipe mismatch, stale mirror) so exception paths are demonstrable and tested. Each simulator arrives in the phase that first needs it (11).

4.7 Security architecture

In plain words: security is designed at the start, so attribution, signatures and records cannot be forged by someone who controls a browser, a badge or the database connection.

- **Threat model** (STRIDE per component) and an OWASP ASVS level 2 design review are Phase 1 exit criteria; Phase 7 keeps penetration testing and hardening verification only (L-56).
- **Sessions:** server-side sessions; Secure, HttpOnly, SameSite=Strict cookies; CSRF tokens; idle and absolute timeouts (kiosk lock default 5 min, allowed 1-15, DI-38); one active shop-floor session per person (a login elsewhere ends the old one); rate limiting on authentication and signing.

- **Device identity:** registered devices hold enrolled device keys or mTLS client certificates; the server derives device ID, class and room from that identity, never from what the browser reports (DI-20). Grade A/B terminals are GRADE_A_B devices bound to their room.
- **Badges and credentials:** secure, non-clonable badge technology (not 125 kHz proximity cards) plus PIN; PIN lockout; badge loss, revocation and replacement with identity re-verification (11.300(c), DI-19); badges tested at issue and periodically; initial and admin-reset secrets single-use and changed before the account can sign, and administrators never see them (11.200(a)(3), DI-15); periodic credential and access recertification (11.300(b), DI-16).
- **Misuse detection:** failed logins and signings, use of locked or disabled accounts and anomalies alert the security role and QA in real time, with tracked acknowledgement and escalation (11.300(d), DI-19).
- **Privileged access:** MFA for the administration console, break-glass and vendor support, and for all remote access (DI-13); vendor remote access is time-boxed, customer-approved and session-recorded, with no GMP rights.
- **Keys and secrets:** on-premise secrets vault with rotation (database credentials, OIDC secrets); seal keys in a KMS or HSM; signing-key rotation with an archive of public keys and algorithm agility so seals remain verifiable for 15 years or more (re-sealing or time-stamping before an algorithm weakens).
- **Application hardening:** strict content security policy, output encoding, schema validation on every input (Zod), dependency and container scanning, signed artefacts.
- **Audit of administrators:** pgaudit forwards to a hash-chained, write-once sink that administrators cannot write.
- **Indian law (verify; not in the inputs):** CERT-In Directions of April 2022 as read from secondary sources: cyber incidents reported within 6 hours, ICT logs kept in India for 180 days, clocks synchronised to NIC/NPL NTP or a traceable source; the incident workflow (DI-12) carries a CERT-In reporting step and the time source is a validated configuration item (5.6). DPDP Act 2023: an inventory of personal fields (eyesight and health declarations, biometrics, Aadhaar-linked eSign references), notice, and a retention justification tied to the GMP rule that requires it; biometric data held only as templates on the device or reader, and eSign held only as provider transaction references (DI-13, IN-09 notes).

---

5. Core logic

5.1 Entities

In plain words: the main kinds of records the system keeps. The kernel part is identical in both editions; the process part has the same shape in both with different content.

- **Kernel and site services (`kernel.*`, `site.*`):** organisation units with time zone; **persons** (verified identity, DI-15) owning **accounts** (unique forever; human, integration or break-glass); credentials, badges, MFA factors, sessions, security events; **registered devices**; **authentication events** (single-use, bound to person, device and purpose); roles, scoped grants and segregation rules; **delegations**; qualifications (training, equipment, gowning, aseptic, eyesight, visual inspection, approved technical staff) with validity; signatures (person, account, legal-name snapshot, role, meaning, delegation, server time, zone, record version, content hash, authentication event, qualification snapshot, status, seal); audit events with hash chains, **checkpoints**, anchors and **restore events**; audit reviews with high-water marks; incidents and DI investigations (DI-12, DI-50); the DI-01 record register; reason codes; attachments; idempotency keys; outbox; time-sync samples; retention rules, dependencies and holds; true copies; configuration and policy versions; control-register snapshot, rule-table versions, market resolutions; material lots, reservations and the batch-number register (site services).
- **Recipe (both editions):** MBR versions (content, hash, change-control reference, scope, per-market authorisation records, validation-basis references, in-flight dispositions); library elements; parameter definitions; block-type registry; where-used index; production orders; batches (pinned MBR, policy and rule versions, issued-record hash, market set, purpose); stage and step executions (loop path, sub-lot instance, attempt, performers); flow tokens and branch decisions; data points (value, unit, channel, reading, entry mode, device, authentication event, `recorded_at`, `occurred_at` with trust level, limit evaluation and version, supersedes link); calculation results; timers and IPC slots; exceptions; deviation links; invalidation records; samples; yields; lots, **output containers**, sub-lots, genealogy links; equipment (PROCESS, INSTRUMENT, UTILITY) and room occupancy; campaigns; line clearances; contingency forms; reviews; manufacturer releases and market dispositions with evaluated checklists; market-add evaluations; post-release links; packaging orders with printed-material reconciliation; shipment lines; evidence epochs.
- **Tablets (`tab.*`):** tablet equipment classes; tooling sets with stroke counts and inspection status; sieves/screens with integrity history; FBD bags with dedication; press streams (side, station, layer); container-level IPC bracketing; printed-tablet QC approval; recovery and reprocessing additions (TB-08); controlled-substance accounts (Decision 14).
- **Injectables (`inj.*`):** clean areas (grade A-D, ISO class, zone type, parent area, requalification dates); EM locations, plans, sessions, results, excursions and their linked batches; gowning, aseptic and visual-inspection qualifications; manual-APS qualifications per operator and configuration; APS runs, envelopes, line status per market; intervention catalogue and log; sterilisation, depyrogenation and SIP cycles with load patterns; filter uses with pre-use and post-use integrity; lyophilisation cycles and loads; hold-time chain (dirty, clean, dried, sterile); WFI draws and samples; inspection results by defect code; CCI results; unit accountability ledger; sterile dossier sections.

5.2 State machines and key guards

In plain words: every record with a life cycle follows an explicit map of allowed moves; nothing can jump outside the map, not even by a direct database edit.

Each machine is defined once as data (XState). From it the build generates the database's allowed-transition table (checked inside the SECURITY DEFINER transition functions), the diagrams, the guard table for reviewers and the model-based test paths. Guards run on the server inside the transaction with the relevant rows locked. Sig(X) = e-signature with meaning X.

General rules for every machine.

- **Time-based validity is computed, not cached** (L-12): every guard computes calibration, qualification, APS, requalification, retest and expiry validity from stored due and expiry instants against the transaction's server

time. Stored time-derived states (CALIBRATION_OVERDUE, EXPIRED) are caches and notifications only. Maximum-duration breaches are computed at every command that touches the step and by a sweeper, so they never depend on a timer job firing (tested with the job workers stopped).

- **Expiry during use:** when an occupancy or step interval crosses an expiry instant (calibration, qualification, APS, clean-area requalification), an exception is raised at that instant. Material status, retest and expiry are checked at every dispensing, issue and charge-in (PR-18); a customer may add checks at each use, never remove these.
- **Involvement** (for S10, 7.2) and every segregation rule are evaluated on the person, not the account (L-02).

5.2.1 MBR version (merges architecture 4.1 and features E.6; per-market effectivity)

- DRAFT → IN_VERIFICATION: author freezes a candidate; content hash computed; static gates pass with zero errors and every warning justified. Any edit returns to DRAFT and voids evidence gathered on the old hash.
- IN_VERIFICATION → SUBMITTED: generated and author-confirmed test vectors pass; sandbox test runs completed on this exact hash with coverage report (or delta verification, 6.5); change record linked; Sig(PREPARED) by the author.
- SUBMITTED → CHECKED: Sig(CHECKED) by an independent checker. **No person who edited the content since it last entered DRAFT may sign CHECKED or APPROVED** (INV-04, L-05); the same rule applies to library elements, RBE rule sets, site values and policy versions.
- CHECKED → APPROVED: Sig(APPROVED) by QA (not an editor, S3; not the checker unless S4 allows); regulatory-conformity sign-off per market (PR-04); a changed Established Condition needs a regulatory assessment with reporting category; **in-flight disposition** for every issued or running batch of the predecessor found by where-used: CONTINUE, HOLD_FOR_ASSESSMENT or CANCEL_UNSTARTED (L-24).
- **Per-market authorisation** (L-17, PR-04): an approved version becomes CURRENT for a market and scope (site, line, batch size) through a signed **market-authorisation record** (QA plus regulatory, the market's approvals or notifications recorded, effective date). It does not change the content hash. When a version becomes CURRENT for a market and scope, its predecessor becomes SUPERSEDED for that market and scope in the same transaction; a successor covering only part of the scope (line 1 only) leaves the predecessor CURRENT for the rest.
- **Bridging validity:** when a change is authorised for one market but still pending for another (e.g., India needs no prior approval, the US supplement is pending), QA and regulatory may give the superseded version a signed, time-limited BRIDGING validity for the first market, usable only by batches whose market set includes a market where the successor is not yet CURRENT, and only while the old process remains within that market's registration. INV-17: at most one CURRENT version per MBR family, scope and market.
- **Selection at issue:** the newest version CURRENT for every market in the set; otherwise the newest version CURRENT or BRIDGING for every market; otherwise issue is refused with an explanation.
- CURRENT → RETIRED by QA (no new batches; running ones per the in-flight disposition); pre-authorisation → WITHDRAWN; returns to DRAFT with reason (signatures invalidated). The first batch after a change needs a post-change evaluation before release (IN-03); PPQ batches carry their flag (PR-09).

5.2.2 Batch record (processing record; each packaging order has its own record with the same states)

- PLANNED → ISSUED: MBR version selected per 5.2.1; batch size one of the approved sizes; ERP order BOM matches the MBR BOM; batch number never used before (voided numbers included) and entered in the allocation register (site services); rules resolved with the table version in force and the MBR's limits not looser than the resolved values (3.3.5); policy versions pinned; for any US market, the 11.100(c) certification record present (5.5); for an export-only product, a valid Indian export permission for this product, strength and destination (status U); injectables: APS envelope checks at issue (5.2.11); control-recipe snapshot and hash stored; contingency pack generated; Sig(ISSUED) "issued, reproduction verified" (PR-10, IN-18, DI-42).
- ISSUED → CANCELLED (nothing started; supervisor Sig + reason; number retired, record kept) or → IN_PROGRESS (first step start, after production's receipt; if the MBR version was superseded or retired since issue, only with a QA signature, L-24).

- IN_PROGRESS ↔ ON_HOLD (supervisor, QA, or automatically on a blocking exception, QA hold point or environment hold; only QA lifts a QA hold). **While on hold** (L-34): no step starts, completes or advances; steps already in progress may still record data and exceptions; the hold type decides whether automated phases must be stopped (a G-06 command); timers keep running; for sterile products, units produced in the hold window are flagged for segregation.
- → ABORTED (production head and QA, two Sig + reason; outputs held or rejected by QA; equipment set dirty). **ABORTED → PRODUCTION_APPROVED → QA_REVIEWED** through an abort checklist: materials accounted for, outputs dispositioned, equipment dirty, genealogy closed, contingency forms reconciled (L-20).
- → EXECUTION_COMPLETE (system): every step terminal, every required signature valid, every stage signed off, yields and reconciliations recorded, every IPC slot done or covered by an exception, every contingency form reconciled (PR-16, DI-43).
- → PRODUCTION_APPROVED: person responsible or designated production reviewer; every exception dispositioned or raised as a deviation; free-text translation complete where the policy requires it (7.4); Sig(APPROVED). Always before QA review (S8). A correction after EXECUTION_COMPLETE returns the batch to IN_PROGRESS; after PRODUCTION_APPROVED corrections go only through a QA reopen (L-09).
- → QA_REVIEWED: every exception-register item dispositioned (7.3); batch audit-trail review signed by a person outside the involvement set, with its high-water mark (5.4); required LIMS results present; reviewed content hash stored; Sig(REVIEWED).
- Any review state → IN_PROGRESS ("reopen" for a correction; QA Sig + reason; review signatures invalidated).
- **Locking** (L-23): a processing record locks at **bulk reconciliation closure**, which needs every packaging order from the bulk closed or the remaining bulk formally held; a packaging record locks at its lot's first disposition. LOCKED → UNDER_AMENDMENT (QA, linked deviation or change record) suspends every affected release (5.2.6) and opens an AP-signed market-impact and recall assessment for each market already released (R-04, IN-29) → LOCKED after re-review and new signatures.

5.2.3 Step execution, corrections and invalidation

- PENDING → ENABLED (token arrives) → IN_PROGRESS (batch not on hold; user holds role and every required qualification now, evaluated on fresh data; equipment, room and material checks pass; time limits respected) → AWAITING_VERIFICATION (mandatory data present, calculations done, minimum duration elapsed, no blocking exception; Sig(PERFORMED)) → COMPLETED (Sig(VERIFIED) or Sig(WITNESSED) by a person who neither performed, entered data in nor corrected the step).
- **Several performers** (L-67): a step may have several performers, each signing PERFORMED for their own entries; the verifier is none of them.
- **Hand-over** (L-07): IN_PROGRESS → ATTEMPT_HANDED_OVER, signed by a supervisor and QA with a reason, when a performer leaves or loses a qualification mid-step. Committed entries stay attributed to the original person and remain valid; a new performer completes the step and signs for their own entries; the verifier covers all of them; the hand-over appears in the exception register. Leaver processing lists open obligations (open steps, pending verifications, owned deviations, delegations) and creates reassignment tasks but never delays disabling the account.
- NOT_APPLICABLE is set only by the system for untaken branches; item-level "N/A" only on items the MBR marks conditional, with reason and signature (PR-16). SKIPPED only for skippable phases, approved by a different supervisor or QA, never for CPP/CQA inputs, reconciliation, line clearance or release signatures. Status is re-checked at commit and at signing, not only at step open.
- **Three different paths for changing what was recorded** (R-18, L-08; DI-24, DI-29, IN-20(c)):
  - **(a) Entry-error correction:** a keying or transcription error in a manually entered value. The corrector re-authenticates with meaning CORRECTED (DI-24), gives a reason, and the S25 approval applies to critical data; the corrector joins the step's data-enterers, so cannot verify it. Device-captured values cannot be corrected, only invalidated. If the original value failed its limit and the corrected value passes, the correction needs attached objective evidence of the true value (device raw string, printout, photo) and QA approval

before any guard, loop exit or release may use it; that approval is the invalidation record. Without such evidence, path (c) applies.
    - **(b) Void of an attempt:** a procedural abort before any result exists (a weighing abandoned before a stable reading, a cancelled label print); reason, attributable signer, QA approval when S28 is on. Once a result exists it cannot be voided.
    - **(c) Invalidation of a result:** only with a linked OOS or deviation investigation that concludes an assignable cause, approved by QA. IPC specification tests follow the OOS workflow (IN-20(c)); the original stays in the record and a passing re-test never replaces it.
    - In every path the exception raised by a failing evaluation is never closed by the change; it must be dispositioned. Every attempt after a failing result is flagged "repeat after fail" in the exception register. Compendial staged tests (weight-variation stage 2, dissolution S2/S3, AQL tightened inspection) are logic inside the block, never voids or repeats (INV-41).
- **Corrections whose dependents were already acted on** (L-09): *derived results* (yields, statistics) are recalculated, the old result is kept and signed dependents are flagged; *decision inputs* (branch condition, loop exit, IPC pass/fail, hold-back) never re-route the executed path: the decision is re-evaluated and, if it would now differ, a "decision-affecting correction" exception only QA can disposition is raised; *executed targets* (a dispense quantity already weighed and charged) are never silently recomputed: a new target outside tolerance of what was executed raises an exception. A correction to a COMPLETED step moves it back to AWAITING_VERIFICATION for re-verification by an uninvolved person and invalidates the stage sign-off.

5.2.4 Exceptions and limit tiers

- **Limit tiers** (L-11, PR-08, E.2), nested alert inside action inside registered (the validator enforces the nesting): **alert** breach → notification and an exception-register item; **internal action** breach → the dependent step stops until a deviation is opened (the MBR may let the current step continue with the deviation raised, PR-08); the deviation is approved by a competent person for its class (PR-13), who is outside the step's involvement set; **registered limit, CQA or CPP** breach → a deviation flagged registration-relevant is mandatory, only QA dispositions it, it appears on every release screen, and the stage cannot be signed off until QA has classified it.
- **Exception machine:** OPEN → DISPOSITIONED (no impact with justification, or deviation raised) by a person outside the involvement set of the step that raised it; in QA review: ACCEPTED_BY_QA; QUERIED → RESPONDED -> DISPOSITIONED; ESCALATED → deviation. Blocking exceptions stop step completion; never deleted; mandatory types cannot be switched off (DI-26).

5.2.5 Deviation (built-in stand-in; with an external QMS the status is mirrored and checked with a version token at release)

- OPEN (linked batches release-blocked at once, or **release-suspended** if already released, 5.2.6) → UNDER_INVESTIGATION (QA classifies minor/major/critical with a risk rationale, IN-02; owner; due date) → IMPACT_ASSESSED (**per lot and market**: none, potential or confirmed, with rationale; QA Sig by a person who is neither the initiator nor the owner) → PENDING_CLOSURE (root cause or most probable cause; "human error" needs a system-factor assessment; CAPA references; regulatory-notification assessment with licensing-authority reference where required; non-empty list of other batches and products assessed with inclusion/exclusion reasons, ST-71) → CLOSED (QA Sig).
- OPEN → CANCELLED only by QA other than the initiator and other than any later release signer of a linked lot, with reason; the cancellation appears on the release checklist (L-21).
- **Deviation-to-batch links are insert-only**; a link ends only with a recorded "assessed, not affected" conclusion and rationale (INV-46). Nobody approves their own deviation (S7).

5.2.6 Release: the India manufacturer release and the market dispositions

In plain words: the Indian authorised person releases each finished lot first; only then can the US or EU decision follow; and a release can be suspended later if new evidence arrives.

- **Manufacturer release** (one per finished lot; India AP, always; Schedule M Part I 11.3.6-11.3.9; IN-05): NOT_READY → READY is computed by the release gate G-REL (5.2.7), never set by hand → RELEASED or REJECTED, Sig by the AP or a named delegate with a valid delegation (checked at signing; S10 applies to the

delegate), after head-of-QC approval, with re-authentication including all components. NOT_READY → REJECTED is also possible (e.g., a sterility failure means READY is never reached, L-20). The evaluated checklist is stored with the signature (IN-05(d)). For a lot packed for India-domestic sale, the manufacturer release is also its India market release.

- **Market dispositions** (one per finished lot and export market; S32 independence holds between markets **except** that each requires the manufacturer release to be RELEASED at that instant, INV-39):
  - **US:** NOT_READY → READY (G-REL plus US items) → RELEASED or REJECTED by the QCU-disposition role. If the AP also holds the QCU permission, **one signing ceremony writes two signature rows**, each with its own meaning, manifestation, content hash and invalidation life cycle (11.50; DI-35 lists QU disposition as its own signature type), replacing v0.1's "one signature, two meanings". A contract giver's QCU disposition is recorded when applicable (PR-54).
  - **EU:** with the in-system QP switch off, once the manufacturer release is RELEASED and the EU items of G-REL are green, the EU disposition moves to AWAITING_QP_CERTIFICATION and the system generates the **EU QP batch package** (R-03): a signed true-copy bundle (DI-41) of the processing and packaging records, the audit-trail review record, a list of every post-entry change to critical data and special observations (EU Ch.4 4.27), deviations with their per-market assessments, the manufacturer release and its checklist, and the manufacturer's batch certificate/CoA required by the technical agreement (Annex 16 content not in the inputs: verify). It is delivered as a package or through a read-only, time-boxed QP account scoped to the lot. The certification comes back as a **structured, signed message** (certifier, MIA number, date, lot) with acknowledgement (DI-39) → CERTIFIED or REJECTED_BY_QP; manual entry is a fallback flagged "transcribed" and verified by a second person (DI-30). With the switch on, only registered QP accounts certify in-system into an append-only register (PR-55).
  - A market whose market-add evaluation has an UNMET_FINAL item is **INELIGIBLE** (terminal, 3.3.3).
- **Cascade:** rejecting, suspending or invalidating the manufacturer release moves every dependent market disposition to RELEASE_SUSPENDED (if released) or NOT_READY in the same transaction; a rejection opens an investigation link visible to every market (PR-56).
- **After release** (L-19): RELEASED (manufacturer or market) → **RELEASE_SUSPENDED automatically** when a deviation, incident, as-found calibration failure, APS failure, EM result, revised LIMS result or DI investigation links to the lot or to any lot in its genealogy after release; the same transaction sends an outbox message that blocks the remaining stock in ERP/WMS and creates an AP-owned recall-assessment task (IN-29). From there: → RELEASED after QA re-confirms with an impact assessment and signature (S10 applies); or → RECALLED; or → REJECTED_AFTER_RELEASE. Suspension propagates through genealogy to downstream lots that consumed the intermediate.
- **Rejected lots** (L-20, IN-05(e)): REJECTED or REJECTED_AFTER_RELEASE → DISPOSAL_RECORDED (witnessed destruction, a linked reprocessing batch, or return); the record cannot close without it; a serialised lot gets a serial-decommissioning task.
- **Shipping** follows INV-42 and the ERP rule in 4.6: nothing ships without a current manufacturer release; EU-destined lots go only to the named importer as quarantined goods until certified.
- **Concurrent release of PPQ batches** (PR-09): by default a PPQ-flagged batch waits for the validation outcome; release per an approved protocol that permits concurrent release, with that batch's own protocol results complete and a per-batch QA signature, is allowed only for markets other than the EU (the India position is not in the catalogs: confirm with the expert).

5.2.7 The release gate G-REL

In plain words: one computed checklist, stored with the release signature, that must be entirely green before the AP (and then each market's authority) can sign.

G-REL is evaluated for a finished lot and market outside the signing transaction and confirmed inside it at an unchanged evidence epoch (5.7). Items, grouped:

1. **Evidence set and review** (L-23(b)): the lot's own packaging record, the bulk processing record and every ancestor intermediate record are QA_REVIEWED with unchanged reviewed-content hashes; no open exception,

non-terminal step or unreplaced invalidated signature; the batch audit-trail review is signed and its high-water mark covers every in-scope audit event, including events on linked records (5.4), or a signed delta review exists (INV-48); late external evidence (EM reads, LIMS results, QMS status, QP reference) is checked by its own items below and shown to the signer as a "since review" list acknowledged item by item.

2. **Deviations and investigations** (L-21, PR-53, R-05): every linked deviation is CLOSED, or IMPACT_ASSESSED with outcome "no impact on this lot for this market" signed by QA who is neither initiator nor owner; "potential" or "confirmed" impact blocks until closure with a disposition; for India every deviation is at least formally assessed (IN-05(c)); cancelled deviations are listed; a regulatory-notification assessment is recorded for every linked change and deviation, with the licensing-authority reference where one is required (Schedule M 11.3.9(e)); additional sampling, inspection and tests for linked changes and deviations are carried out or, where the AP records why that suffices, initiated (11.3.9(f)). **Open-investigation path** (only when the PR-53 switch is on): an e-signed justification by the release authority, never for reconciliation discrepancies, registered-specification failures or, in the Injectables edition, any sterility-assurance investigation (Grade A growth, EM or PM action excursion, APS failure, filter integrity, sterilisation/depyrogenation/SIP, CCI, sterility or endotoxin), which must be closed with a batch-impact disposition (ST-21) or formally concluded as not affecting this batch (ST-71).
3. **Testing and samples:** release tests passed with head-of-QC approval (IN-05; 11.3.9(i)); all samples taken, including reserve samples (PR-44, IN-32); QC decision on extra testing for reprocessed or recovered material (PR-47); no LIMS result revised since review without a re-review.
4. **Reconciliation** (L-22, IN-17, PR-39): yield, printed-material and serial reconciliation within limits, **or** outside limits with a CLOSED investigation whose QA-signed conclusion explains the discrepancy quantitatively and, for printed materials, rules out mix-up (100 % inspection or destruction of the implicated units per SOP). Any correction of a counted quantity after its evaluation follows path (a) or (c) of 5.2.3.
5. **Materials** (L-23(a)): every consumed lot was usable at consumption (stored evaluation), and since consumption no consumed lot has become REJECTED, RECALLED or ON_HOLD or had a QC result revised. A lot expiring after it was correctly consumed does not block.
6. **Licence, change and validation:** licence and product-approval conformity for this product, strength, pack and market (11.3.9(a)); the MBR's validation-basis references for principal manufacturing and test processes present (11.3.9(c)); post-change evaluation done (IN-03); PPQ outcome or the concurrent-release rule (5.2.6); export permission valid for an export-only product (U).
7. **Incidents and alarms:** no unassessed incident or critical alarm (DI-12, DI-49); no open DI-investigation hold (DI-50).
8. **Market-add evaluation:** no UNMET item (3.3.3).
9. **Finished-pack examination** recorded (Schedule M 11.3.7; TB-24 or the injectables packaging examination).
10. **Edition items:** Tablets: printed-tablet QC approval (IN-37), held containers dispositioned (5.2.9). Injectables: every sterile-dossier section signed (ST-70); every contributing sterilisation, depyrogenation and SIP cycle reviewed and conforming (ST-72); every EM and PM result in (no plate incubating) and every excursion assessed, with EM linkage per 5.2.10 (ST-24); APS valid for the whole fill and the batch not inside an APS-failure scope (5.2.11); sterility and endotoxin results (ST-65); all filter integrity results (ST-52); 100 % inspection, CCI and accountability (ST-62, ST-64, IB-25); every unlisted intervention dispositioned (ST-36).
11. **Market items:** US: release-rate result per active for controlled-release tablets (PR-54), micro testing where required, sterility and pyrogen tests, contract-giver QCU disposition where applicable; EU: QP package generated; each market's registration conformity and its market-specific packaging record complete (PR-56).

5.2.8 Equipment, instruments, rooms and lines

- **Equipment kinds** (L-27): **PROCESS** equipment has exclusive occupancy, a cleanliness region and campaign counters; **INSTRUMENT** (IPC balance, hardness tester) has no occupancy and no cleanliness state, only calibration and a per-use check, so one tester can serve two presses; **UTILITY** (WFI loop, HVAC) takes its status from outside feeds. INV-12's no-overlap rule applies to PROCESS equipment only.
- **PROCESS equipment regions:** *cleanliness* CLEAN → IN_PRODUCT → DIRTY (dirty-hold clock) → CLEANING → CLEAN (manual cleaning: performer plus different verifier; validated CIP: one verifier, S22; clean-hold clock

starts); cleaning levels MINOR and MAJOR; **campaigns** of the same product with minor cleaning between batches, within maximum batches and maximum days with a validation reference (PR-27, IN-26); injectables add DRIED and STERILISED with hold limits and re-sterilisation of product-contact items every batch unless a validated campaign exists (ST-38). *Fitness*: QUALIFIED, CALIBRATION_DUE_SOON, CALIBRATION_OVERDUE, OUT_OF_TOLERANCE, MAINTENANCE, OUT_OF_SERVICE, RETIRED (computed per the general rule); an as-found calibration failure lists every batch since the last good calibration, opens a linked deviation and suspends released lots among them. *Usage*: IDLE or IN_USE(batch). Use guard: clean within hold at the required level, qualified, calibration valid for the planned interval, idle; tablets: tooling inspected and within stroke limit, sieve checked; injectables: sterile items within sterile hold. Instruments are re-checked at every reading.

- **Room / line:** DIRTY → CLEANING → CLEAN → CLEARED_FOR_BATCH → IN_USE → DIRTY; OUT_OF_SERVICE; rooms may contain **sub-locations** (dispensing booths) with their own status. The **clearance type** depends on what came before: product change (full clearance and cleaning), same product next batch in a campaign (campaign clearance), and, for packaging, **any change of packaging order** (L-29): clearance is mandatory between any two packaging orders, keyed on packaging order and artwork set, even when both come from the same bulk batch; the checklist includes removing and reconciling the previous order's printed materials and serial pool; the printer job, vision recipe and serial pool are bound to the packaging order; a batch-data correction after printing holds everything printed since the change. Clearances are performed and, where required, verified by a different person (S23, S24), with previous product, batch and order recorded. One active packaging order per line (PR-35). Area logbooks are written automatically for every room (3.3.6).

5.2.9 Material lots, output containers, reprocessing and recovery

- **Material lot:** QUARANTINE (incl. on test) → RELEASED (tests passed, CoA reviewed, approved supplier; QC/QA Sig) with a recorded **release route** (identity-tested, or India-only vendor-history route under Schedule M 14.15 when QA has enabled it with a statistical justification) or REJECTED; → RELEASED_FOR_FURTHER_PROCESSING (intermediates, QA hold point); RELEASED ↔ ON_HOLD (QA); RETEST_DUE and EXPIRED computed by date; RETURNED and RECALLED by QA; CONSUMED at zero; DESTROYED only with a witnessed record. A finished lot leaves quarantine only through the release model (5.2.6). **Consumption guard:** right item; usable status; within retest and expiry; container dispensed for this batch (scan); reservation and quantity available; hold time not exceeded; **release route satisfies the consuming batch's resolved identity-test rule** (R-09).
- **Output containers** (L-30): OPEN → CLOSED; CLOSED → HELD → SORTED, REJECTED or RELEASED_BY_QA (QA signature). An IPC failure or missed check holds every container whose open interval overlaps (last passing check, failing check]; per stream only where streams discharge to separate containers, otherwise the whole press output. Held containers cannot be merged, consumed or packed (TB-11).
- **Reprocessing and recovery** (L-28, TB-08, K-24, IN-19, PR-47), four distinct cases:
  1. **Whole-batch reprocessing or rework of non-conforming material** (K-24/G-24): QCU authorisation citing an investigation, an approved reprocessing MBR, a risk evaluation, a **new batch number** linked to the parent(s) (INV-33), QC decision on extra testing, product-level disable where the registration forbids it.
  2. **Reprocessing addition** (TB-08 class b): rejected in-process material of the same product (e.g., milled compression set-up or weight-control rejects) added into a later batch at an MBR-defined step, with an approved written reprocessing procedure, QCU approval before execution and a risk evaluation.
  3. **Recovery** (TB-08 class a): conforming material from an earlier batch added into a later batch at an MBR-defined step, with prior authorisation and a risk evaluation addressing shelf life.

  - For 2 and 3 the receiving batch keeps its own batch number; the source is a container or lot created in an executed batch record (residue generated at inspection becomes such a container, with its generation date, quantity and a status of "reserved for reprocessing" in quarantine or, for recovery, released); maximum percentage, maximum age and maximum number of source batches are MBR limits; chained recovery is blocked; the receiving batch's expiry is no later than its own product rule and the earliest expiry among recovered inputs, with QA confirmation (IN-19(b) default proposal); a QC decision on extra testing is mandatory before release.

  4. **Validated re-treatment inside the process** (LOD re-dry, continued coating to weight gain) is a bounded loop (6.2), not rework.

- The sample BMR's "rework addition, at most 3 % of batch and 3 months old" is modelled as case 2, because its source is rejected material (TB-08 note); the MBR author must declare the case, and the validator refuses an addition step without one.

5.2.10 Injectables: clean areas and environmental monitoring

- **Clean area:** CLEANING_DISINFECTION → CLEAN_READY (cleaning verified; isolators: passed bio-decontamination and glove tests) → IN_OPERATION (validity not expired; requalification including the final-filter leak test within 6 months for A/B and 12 months for C/D in every profile, ST-14, 3.3.6; pressure and HVAC within limits from the BMS; EM session started; every entrant qualified for the grade and reconciled against access-control entries or gowning-room badge-in, L-33; APS valid and planned interventions covered) → DIRTY; any → ENVIRONMENT_HOLD on a non-viable action alarm or pressure failure beyond its allowed time; → REQUALIFICATION after shutdown, construction, a final-filter change or an HVAC change. Areas form a **hierarchy** (a Grade A zone inside its Grade B background room; isolator and RABS mapping per ST-74).
- **EM result:** RECORDED → ALERT_REVIEWED, or EXCURSION_OPEN → deviation → IMPACT_ASSESSED → CLOSED. **Linkage** (R-13, L-31, ST-21, ST-22): an **action excursion in any grade** links every batch processed in that room or zone **or in any zone inside it** (a Grade A zone inherits excursions of its Grade B background) between the sampling time and the time the result was reported; linked unreleased batches go ON_HOLD and released ones are suspended; any **Grade A/ISO 5 growth** (no alert tier, action at any recovery) also links every batch whose aseptic window overlaps the sample, and release waits for closure with a documented batch-impact disposition; a **personnel-monitoring** alert or action links every batch the person worked in that session; a **utility** excursion (WFI, clean steam) links every batch that drew from that point in the window. Grade A/B isolates are identified to species (ST-21). Each result is evaluated individually, never only as an average (IN-42).
- **Completeness** (ST-18): planned versus taken; a missed or late sample is an exception that blocks release until assessed. **Adverse trends** (ST-20 trend rules, each required with configurable parameters) create an excursion event linked to every batch in the affected area and period.

5.2.11 Injectables: APS status, APS record, interventions and operator qualification

- **APS status** per aseptic process × line × shift × container-closure configuration (ST-29, IN-45, Annex 1 9.38-9.49; R-14): NOT_QUALIFIED → INITIAL_VALIDATION_IN_PROGRESS (counter of consecutive successful runs covering every shift the process runs; at least 3) → QUALIFIED → DUE → EXPIRED. **Expiry** (rule key `sterile.aps.periodicInterval`, L-68(g)): the same site-local calendar day six calendar months after the fill end of the last successful APS (last day of the month when that day does not exist), at 23:59:59 site time, with no grace. QUALIFIED → **REVALIDATION_REQUIRED** on a change flagged as affecting sterility assurance (HVAC, equipment, process, number of shifts or personnel, major shutdown), a new container-closure combination, or inactivity longer than the site parameter; FDA triggers (significant staffing change, EM anomaly, extended shutdown, finished-product sterility positive) move it to APS_ASSESSMENT_REQUIRED, which QA may clear with a documented risk decision except where the change affects the aseptic process on a line under the EU rule (9.49). A failed or invalid APS → SUSPENDED: production suspended; everything made on the line since the failure quarantined; all batches since the last successful APS listed for per-batch record review and sterility-breach assessment, unreleased ones release-blocked and **released ones suspended for a market-impact decision**; root cause, CAPA, species identification; → REPEATS_IN_PROGRESS (normally at least 3 consecutive successful repeats under the EU rule) → QUALIFIED; implicated operators restricted (ST-33). A separate status tracks the formulation/bulk aseptic stage of non-filterable products (at least twice a year, ST-29). Under Decision 2 option B "any contaminated unit fails" applies on every line; under option A the FDA/India tiers apply only on lines with no EU-market and no PIC/S-destination product (R-28).
- **Envelope** (ST-31, IB-15; R-15), derived only from successful APS records: container sizes, APS unit count, maximum duration, line-speed range, maximum holds of exposed product and equipment, peak headcount and operator time in the cleanroom, interventions with maximum counts and durations, idle interruptions, shifts, campaign length, lyo items (SIP to use, filtration to loading, trays, loading duration). Checked **at issue** (container size covered; for an APS under 5,000 units the batch is not larger than the APS), **at fill start** (line, shift, format, operator participation; APS valid through planned fill end plus a margin), **continuously during the fill** (duration, headcount, intervention counts, speed, holds, lyo loading time) and **at fill end**. Any exceedance raises a deviation that blocks release until QA assesses it.

- **APS record** (V-10, IB-28, ST-32; L-35): ISSUED → IN_PROGRESS → INCUBATING (FDA default scheme enforced: 20-35 °C within ±2.5 °C of target, at least 14 days in total, at least 7 days at each of two temperatures with the lower first; labelled FDA, since Annex 1 only requires justified, validated conditions) → READ (interim and final reads by qualified readers with a mandatory QC-observer signature) → ACCEPTED, FAILED or INVALID (QA signature; invalidation only by an SOP rule equally applicable to commercial lots). Side effects in the same transaction: line status, envelope, operator APS participation, ST-33 look-back. It can never be released and consumes no product inventory. INV-44: filled = incubated + justified non-incubated + separately incubated set-up units, with zero unexplained; all integral units incubated, including cosmetic rejects; removal without incubation never above the catalogue number or above production.
- **Interventions** (ST-34 to ST-36; L-32): the quick-log offers catalogue entries qualified in a valid APS for the line **and always an "Unlisted intervention"** with description and position, which raises a critical exception and a deviation, needs real-time QA authorisation before the operator continues (remote with MFA allowed; S30), flags the units and window for segregation per SOP, and blocks release until assessed. The catalogue limits what is allowed, never what can be recorded. Occurrences above the APS-demonstrated count, or a rare intervention not simulated within its interval, raise a deviation.
- **Operator qualification:** NOT_QUALIFIED → IN_TRAINING → QUALIFIED → EXPIRING → EXPIRED; QUALIFIED → SUPERVISED_ONLY / RETRAIN / SUSPENDED (failed gown or glove monitoring, failed APS, deviation) → QUALIFIED only with requalification evidence; any → REVOKED. Unsupervised Grade A/B signatures need gowning qualification within 12 months, current aseptic training and a successful APS within 12 months on this line covering the activity (ST-08). **Manual aseptic processes** (ST-30): a separate qualification per operator × container type × closure × equipment train, three consecutive successful APS initially and one about every 6 months per operator, checked when a manual aseptic step is signed. The status found is stored with each signature.

5.3 Invariants that must always hold

Each has an application guard, a database enforcement (constraint, trigger, privilege or SECURITY DEFINER function), a bypass test that attacks it with raw SQL under the application's role, and an invariant query that must return zero rows; production runs the same queries on a schedule. Section 9.1 states which attacks are prevented in the database and which are detected afterwards.

ID	Invariant
INV-01	A GMP batch has exactly one pinned, immutable MBR version that at issue was CURRENT or BRIDGING for every market in its set and in scope
INV-02	The issued control recipe never changes; its stored hash matches
INV-03	A verifier or witness is never a person who performed, entered data in or corrected that step
INV-04	No person who edited a version's content since it last entered DRAFT signs CHECKED or APPROVED; author, checker and approver are distinct persons (checker = approver only if S4 allows)
INV-05	At the instant of any manufacturer release or market disposition, G-REL held for that lot and market at an unchanged evidence epoch
INV-06	Data points, calculations, genealogy links, signatures and audit events are insert-only; corrections add linked rows with a reason
INV-07	Every GMP insert or update writes an audit event in the same transaction (user, person, role, server time, zone, old/new or row hash, reason for changes)
INV-08	Every audit chain verifies from its last checkpoint; anchors match, or a signed restore event covers the gap
INV-09	A signature is valid only while its content hash is unchanged; a change invalidates it in the same transaction
INV-10	Every signature consumes one unused authentication event of the same person, bound to its challenge, with the components the policy requires; signatures are inserted only by the signing function

ID	Invariant
INV-11	No person holds an administrator role and a GMP role anywhere in the installation; at most one active GMP-capable account per person; accounts are never deleted or reused
INV-12	PROCESS-equipment occupancy never overlaps; equipment and instruments are used only clean (process), qualified and calibrated; crossing an expiry during use raises an exception
INV-13	Consumption only from usable lots with a satisfying release route and a reservation; lot quantity never negative
INV-14	Every consumption and output creates genealogy links in the same transaction
INV-15	A stage or packaging order starts in a room/line only after a passed clearance of the required type recorded after the previous occupancy or order
INV-16	Every event's `recorded_at` comes from the database clock; any claimed or source time is stored separately as `occurred_at` with a trust level
INV-17	At most one CURRENT version per MBR family, scope and market; BRIDGING validity only through a signed, time-limited market-authorisation record
INV-18	Test, simulation and training data live in a separate database; GMP batches only from CURRENT or BRIDGING versions
INV-19	Loops and replicate sections never exceed their maximum without an exception; all iteration data kept
INV-20	No step completes before its minimum duration; every maximum breach creates an exception, computed at every command and by a sweeper
INV-21	A locked record changes only through the amendment process; a processing record locks only at bulk reconciliation closure
INV-22	Nothing is purged before its retain-until (including the maximum over its dependants) or while on hold; purge only through the retention function, which leaves a tombstone and the chain checkpoint
INV-23	Injectables: aseptic filling only in an area IN_OPERATION with an active EM session, APS valid through planned fill end and qualified operators; release waits for all EM results and for every linked sterility-assurance investigation to be closed or concluded no-impact for this batch
INV-24	Tablets: compression only with inspected tooling within stroke limits
INV-25	Derived values (expiry, theoretical yield, scaled and potency-adjusted quantities, counts, yields) are computed, never typed
INV-26	Every signer and data-enterer held the required role and qualification at that instant, evaluated on data within its freshness bound; the evaluation is stored
INV-27	Batch numbers, contingency-form numbers, user IDs and signature IDs are never reused, including after a restore
INV-28	India is in every market set; for every atomic rule key the resolved value is at least as strict as India's
INV-29	Every rule evaluation stores value, governing market, predicate branch and rule-table version
INV-30	Production approval precedes QA review
INV-31	No verification by zero humans; single-person automated checking only on the step types 21 CFR 211.68(c) covers (S5a)
INV-32	Labelling and printed-material reconciliation always runs; no waiver for Indian-made product
INV-33	Whole-batch reprocessing produces a new batch number linked to its parents; every reprocessing or recovery addition is linked by genealogy with source batch numbers and quantities
INV-34	An MBR cannot become CURRENT unless every intermediate has a hold time, every timer and yield checkpoint has limits and a source, and (tablets) the IPC interval meets `tablets.compression.ipcSampleMaxInterval` with periodic station-

ID	Invariant
	wise checks on multi-point presses
INV-35	The batch audit-trail review, the manufacturer release and every market disposition are signed only by persons outside the batch's involvement set
INV-36	Injectables: pre-use filter test always recorded; post-use test before removal from the housing; redundant train: backup tested if the primary fails; serial train: every filter passes
INV-37	Injectables: a passing sterility test never overrides a process failure (ST-68)
INV-38	GMP decisions never read a projection; they read system-of-record tables, and external mirrors only within their freshness bound
INV-39	One market's disposition never changes another's, except that every market disposition requires the lot's manufacturer release to be RELEASED at that instant, and suspending or rejecting it suspends the dependants in the same transaction
INV-40	Every GMP command from a SHARED device carries a fresh authentication event of the person it is attributed to
INV-41	A failing evaluation never becomes passing (by correction, void or new attempt) without an approved invalidation record; the exception it raised is closed only by disposition
INV-42	No shipment of a lot without a current manufacturer release, or with an open post-release link; EU-destined lots only to the named importer in quarantine until certified
INV-43	A market disposition cannot become READY while its market-add evaluation has an unmet item
INV-44	Injectables APS record: filled = incubated + justified non-incubated + separately incubated set-up units; removal never above the catalogue number
INV-45	Injectables: fusion-sealed containers of 100 mL or less have 100 % CCI; visual inspection is never a CCI method; vacuum-sealed containers pass vacuum retention before release (ST-64)
INV-46	Deviation-to-batch links are insert-only; a link ends only with an "assessed, not affected" record
INV-47	No policy loosening applies to a batch issued before the loosening's effective date
INV-48	At disposition, the audit-trail review's high-water mark covers every in-scope audit event, or a signed delta review exists

5.4 Audit trail

In plain words: the database writes a tamper-evident diary of every change, and nothing can switch it off; closed records are sealed and archived so the diary stays verifiable for decades.

- **Capture:** row-level triggers on every GMP table write field-level entries in the same transaction; the application sets the actor context (person, account, role, session, device, authentication event, reason, correlation ID, source) and a missing context fails the transaction. For insert-only tables the audit event references the row hash instead of copying the row. The domain layer adds a plain-language business entry with the same correlation ID ("weighing recorded, 12.503 kg, balance B-17, container C-0042").
- **Content:** who (person, account, legal name, role), what (record, field, old/new), when (`recorded_at` in server UTC + site zone and offset; `occurred_at` and trust level where relevant), why (reason code and text, mandatory for every change or void after first entry), how (UI, API, integration, system; device). Creation is audited (DI-22).
- **Always on** (DI-23): no setting disables, pauses, filters, edits or purges it; a start-up and hourly check confirm every GMP table's trigger is enabled, otherwise GMP writes are refused and an alarm raised.
- **Tamper evidence:** per-aggregate hash chains (batch, MBR version, lot, equipment, room, person, configuration area); every few minutes the chain heads form a Merkle root written to write-once storage (optionally an RFC 3161 time-stamp). At record closure a **chain checkpoint** (head hash and count) is sealed. Verification runs nightly over everything changed since the last checkpoint, with a rolling full verification over a configurable number of

weeks (L-46). A superuser who rewrites rows still cannot match the anchors, and pgaudit records the session. A restore is accepted only through a signed restore event (4.2).

- **Archival and purge** (L-46, DI-46, DI-47): closed records are exported as verified true-copy packages (canonical PDF/A-3 rendering plus JSON, signatures, audit trail, manifest) to write-once archive storage with periodic fixity and readability checks. Purging after retain-until is done only by an approved retention job through a SECURITY DEFINER function that leaves a tombstone and the checkpoint, so remaining chains still verify.
- **Review** (R-17, L-10, DI-25): the **batch audit-trail review report** covers (1) the batch chain and every packaging-order chain; (2) **linked-record changes**: audit events on every lot consumed, every piece of equipment and instrument used, every room, every involved person's qualifications and role grants, the EM/PM results, sterilisation cycles and LIMS results the batch relied on, and the rule-table version, site values, policies and MBR version in force, from the last release, clean or qualification of each object before its use up to the review; (3) batch-linked system events (interface failures, signature failures, lockouts, time-sync faults, alarms). Each guard evaluation stores the audit-event ID of the last change to each field it read (retest date, calibration due date, qualification validity), and the report flags as critical any change to an evaluated field within a configurable period before use (default 72 h). Criticality orders and highlights but never hides a post-entry change. The review stores the report version, its hash and a **high-water mark** (the last audit-event ID per included chain); any in-scope event after the mark needs a signed delta review before disposition (INV-48; DI-25 test: correcting a cleaning-log entry of used equipment after QA review keeps disposition blocked until the delta review). The reviewer is outside the involvement set (INV-35).

5.5 Electronic signatures and attribution

In plain words: the server shows exactly what you sign, you prove it is you at that moment, and the signature is mathematically tied to that content; on shared terminals every single entry, not just signatures, belongs to the person who just authenticated.

1. **Challenge:** the client sends record, meaning and the hash of what it displays; the server rebuilds the canonical content (values, units, limit evaluations, performers, MBR version, attachment hashes), checks the hash, and returns a single-use challenge valid for 120 s with the manifestation preview.
2. **Authenticate:** the authentication service verifies the components the policy requires and inserts a single-use authentication event bound to person, device and challenge (4.4).
3. **Sign:** the signing function consumes that event, runs the guards (role, delegation, segregation on the person, qualification, performer is not verifier, record state), inserts the signature row (content hash, authentication event, name, role, meaning, server time), changes state, writes audit and outbox rows and seals the row with a server key in a KMS/HSM. Any failure aborts everything. A seal-verification job flags any signature whose seal does not verify.

- **Components and sessions** (R-19; 11.200; DI-38): two distinct components (badge + PIN, or user ID + password). Default: all components at every signing. The QA-approvable relaxation applies only to routine signings by the **session owner** within one continuous session, and then leaves the **private** component (PIN or password), never the badge or ID alone; it never applies to disposition, certification or MBR approval. A signature by anyone other than the session owner (a verifier at the performer's kiosk) always requires all components. The continuous session ends at lock, inactivity timeout, logout or change of terminal. Biometric sign-in on kiosks is allowed; biometric **signing** only with evidence that it cannot be used by anyone but its owner, including liveness (11.200(b)); otherwise biometrics are sign-in only. Test: a verifier signing in the performer's kiosk session with PIN only is refused.
- **Identity-provider step-up** (L-54): OIDC step-up uses `prompt=login` and `max_age=0`; the server checks `auth_time` is after the challenge was issued, a nonce bound to the challenge, and `acr` / `amr` values that meet the signature policy; on shared devices each signing uses an isolated IdP session that is cleared afterwards. A test with a live IdP single-sign-on session proves the signature is refused.
- **Attribution of every entry on shared devices** (L-01; features U10; DI-14): on SHARED and GRADE_A_B devices every GMP command (value, scan, correction, exception, void) carries a fresh identity assertion bound to it in the same transaction: a badge tap within the last N seconds (default 60, set per device class) or a badge-present

session in which removing the badge locks the device; the data point stores its `authentication_event_id` (INV-40, enforced by a constraint). Extending a session before the inactivity lock needs a badge tap on shared devices. A bypass test inserts a data point from a shared device without a valid authentication event: refused.

- **Meanings** (DI-35): a closed catalogue of every signature point a rule requires; customers can add points, not remove the minimum. **WITNESSED** means the signer watched the act: it can be signed only on the same device or a device registered to the same room, within a configurable window (default 5 min) and before the next step starts; a late witness raises an exception. **VERIFIED** means review of recorded evidence and may be signed later; the validator requires WITNESSED wherever a critical value has no objective evidence to verify (a manual reading without device capture, scan or photo) (L-03; PR-21's room-terminal window for charge-in stays a switch, default off). Other added meanings: CORRECTED (DI-24), RE-VERIFIED AFTER CORRECTION (a qualified, uninvolved person replaces an invalidated signature whose signer is unavailable, never in the original signer's name, L-07), and per-market disposition meanings. Manifestation: legal name, user ID, role, meaning, date/time with offset and zone on every display and print (DI-36). Any change to signed content (including a recalculated dependent) invalidates the signature in the same transaction; it stays visible and a new signature is required (DI-37). No "remember me", pre-signing or proxy signing; an India AP delegate signs with their own credentials and a cited, valid delegation (S16, IN-05(b)).
- **Part 11 administration** (R-20): identity verified before a signing account is activated; first-use acknowledgement; lockout after 5 failed attempts by default (3-5 allowed); secrets, badges, misuse alerts and recertification per 4.7. The **11.100(c) certification record is a tenant prerequisite whenever any batch has US in its market set** (it binds the Indian exporter whose records support US batches, not only "US tenants"); US-market batches cannot be issued without it (a `proposed_modify` on DI-15, which only flags).
- **India option** (IN-10, Decision 3): a second signature for selected meanings with a Class 3 DSC or eSign over the same content hash; off until a licensed CA or eSign provider is configured and its chain validates to the CCA root in OQ; India-licensed tenants see a banner that counsel questions D.5 Q1-Q3 are open.
- **Signature workload** (L-63): each golden batch has a **signature budget** used as a UX acceptance metric; one PERFORMED signature covers one IPC slot's readings (group signing within one signature point, never across points a rule requires separately); badge tap + PIN keeps each signing short; the DI-38 relaxation remains a QA decision rather than the default.

5.6 Time

- **Two times per event** (L-13, DI-27, DI-28): `recorded_at` from the database server clock is authoritative for ordering and audit; `occurred_at` is the claimed or source time with a trust level: DEVICE_SYNCED, DEVICE_UNSYNCED, USER_CLAIMED or PAPER. Time constraints and EM-window linkage use `occurred_at` when its trust is DEVICE_SYNCED, otherwise `recorded_at`, and record which was used. `occurred_at` must lie between the step start and `recorded_at` and be in order within the step. A late entry evaluates status guards at both times and raises an exception if either fails; it never blocks silently and never passes silently; beyond the late-entry window it needs a supervisor co-signature. The late-entry window is a kernel rule key with edition defaults (kernel: end of the planned step window; injectables: 15 min).
- **Storage and display:** UTC with the site zone; displays show site time with offset regardless of browser; durations use UTC instants, so daylight-saving changes cannot distort them. **Calendar rules** (L-40) use the site-local calendar day: manufacturing date, "daily" checks, shift boundaries. Each product names its **shelf-life start event** (e.g., first dispensing of the API or first mixing; PR F-9); expiry follows the product rule (e.g., the earlier of start + 36 months and the API lot's expiry, as in the sample BMR), never later than the earliest recovered-input expiry (IN-19(b)), rounded by the product's convention (e.g., end of month), computed never typed.
- **Drift and monotonicity** (L-14): a monitor records the offset of every database node against at least two independent references; above the alarm threshold (default 1 s, allowed 0.1-5 s) it raises a data-integrity incident and flags entries in the drift window. It does not block capture (DI conflict 17); the contingency path exists instead. A `recorded_at` earlier than the aggregate's last value is flagged and ordered by the per-batch counter; the failover runbook includes the offset check. The **time source** is a validated configuration item recorded with every drift sample, defaulting for Indian sites to an NPL/NIC-traceable NTP service (R-33, verify).

5.7 Concurrency and idempotency

- Optimistic locking on every record (stale commands receive the current state); short row locks in a fixed order (batch, step, lot, equipment, chain head) for rules spanning records; exclusion constraints for no-overlap rules. Every mutating request carries an idempotency key, so double clicks and network retries are harmless. Several operators can work one batch in parallel branches; locks are per step.
- **Evidence epoch for release** (L-45): each lot and batch has an `evidence_version` incremented in the same transaction by any write that can affect release (exception, deviation link, lot status, LIMS result, EM link, correction, audit event in review scope). G-REL is computed outside the signing transaction, with external status (QMS, LIMS) fetched beforehand with version tokens. The disposition transaction locks the disposition row and the epoch rows (`FOR UPDATE`), checks the epochs and tokens are unchanged and still fresh, and commits under READ COMMITTED; a writer arriving after the lock waits and, once released, triggers the post-release path. MBR authorisation and QP certification use the same pattern. No external call runs inside a transaction.

5.8 A running batch is never affected by later changes

The batch pins its MBR version; the issued record is a hashed snapshot of the resolved, scaled master; every internal reference (library elements, parameter definitions, material and equipment-class versions, SOP revisions, templates) is pinned by version and hash, never "latest" (DI-10). The rule-table versions, resolved values and **policy versions** are stored at issue; the RBE rule-set version is fixed at production approval (L-25). A newer table or policy applies to a running batch only as a tightening or through a QA change-control decision that triggers an audited re-resolution. The batch records the software release it ran on (DI-03).

Block versions (L-49): each block is split into a versioned **execution handler** (retirable once no open batch uses it) and a **record model and renderer**; at record closure a canonical rendering plus JSON is frozen into the archive package, so closed records never need old execution code to be read for their retention period. Block data carries versioned schemas with upcasters, and every block version ships a configuration migrator with tests. **Upgrade policy:** an upgrade may proceed with batches open when its semantic-change report lists no handler that an open batch uses; otherwise the change ships as a new block version used by newly issued batches. A security fix that changes semantics of a handler in use needs a QA impact decision per open batch.

5.9 Disagreements between inputs and their resolution

1. **MBR state names** (architecture vs features E.6): merged into 5.2.1, keeping all gates and the three signatures; effectivity is per market (PR-04).
2. **Kernel block numbering** (K-01..K-26 vs KB-01..KB-19): one canonical list G-01..G-25 with both aliases (6.6).
3. **Batch "RELEASED" with an in-system QP** (architecture 4.2) vs no QP slot at the Indian site (features A.6, India correction 8, PR-55): manufacturer release plus per-market dispositions; external QP reference by message; in-system QP only by switch.
4. **Markets independent** (PR-56, S32) vs Schedule M 11.3.8 (no sale or supply before AP certification): independence holds between export markets; the India manufacturer release gates all of them.
5. **"Uncertified stock cannot be shipped"** (DI-39 wording) vs EU certification after importation (PR (b)): read as "cannot be moved to saleable status"; quarantine shipment to the named importer is allowed.
6. **Area logbooks "EU in set"** (PR D-14), **specimens "US in set"** (PR D-1) and **processing checklist configurable** (PR D-5) vs India's always-applicable rules (Schedule M 17.3.10.17 P~, IN-14, IN-21): India wins; all three are always on.
7. **S10** switch (outline v0.1 3.5) vs LOCKED (features A.5): locked; only the QA-in-process question is a switch.
8. **S17** configurable (features A.5, IN-04) vs every batch India-licensed (assumption 1): locked; register content configurable.
9. **11.100(c) record** flagged only (DI-15) vs refuse US issue without it (R-20): refuse, proposed as a MODIFY for the expert.
10. **Action-limit breach handling** (critique L-11: supervisor exception) vs PR-08 (deviation): PR-08 wins.

11. **Blocking signatures on clock drift** (architecture 5.4) vs alarm-flag-contingency (DI-27, DI conflict 17): DI catalog wins.
12. **Offline signing on devices** (features extra 12) vs "terminals never sign" (architecture 5.6): not in v1 (Decision 12).
13. **Re-authentication** always both components (architecture) vs a QA-approvable relaxation (DI-38): strict default, relaxation as a switch limited to the session owner's private component.
14. **Default sterile market profile** "EU + US" (ST catalog) vs "India + EU + US" (injectables spec): India always present; line-level defaults follow Decision 2.
15. **Stage frame K-02** exists only in the Tablets spec, but IN-15 applies to both: generic block G-02.
16. **Late-entry window** defaults differ (kernel vs injectables 15 min): one kernel rule key with edition defaults.
17. **Requalification interval for India-only A/B rooms** (B-16: warn 6, block 12) vs ST-14 (6 months): 6 months in every profile as a product choice.

---

6. MBR designer and building blocks

6.1 Structure (ISA-88)

In plain words: a master record follows the industry's recipe standard, from the whole procedure down to single steps.

Procedure (one MBR per product, strength, batch size and, for injectables, line) → **stage** bound to a room, line or equipment train → **operation** (e.g., "dry to LOD endpoint", "PUPSIT") → **phase** = one configured block. Each stage runs in a stage frame (G-02) that records start and end, rooms and equipment with previous product and batch, the person responsible (IN-15), stage yield, and a derived "completed per MBR / with deviations" statement nobody can type. Packaging is a separate procedure per packaging order, linked to the bulk batch (PR-01, IN-14).

6.2 Block-structured flow

In plain words: the designer draws a process only with shapes guaranteed to finish properly; there are no free "go to" arrows.

- **Sequence; parallel** branches with an all-must-finish join; **alternatives** decided by recorded data, by an authorised operator's choice with reason, or at issue for variants (untaken branches are marked not applicable by the system); **repeat-until loops** with a mandatory maximum (and optional duration) that raise an exception when exhausted, allowed only on process endpoints (LOD, pH, weight gain), never on release or acceptance tests ("re-blend until uniformity passes" is refused; that is reprocessing, G-24); **hold points** that lock everything downstream until QA signs; **library calls** to pinned element versions.
- **Replicate per sub-lot** (L-26): a bounded section repeated for granulation parts, drying loads, coating pans, autoclave loads, lyophiliser loads or packaging part-batches. The count comes from an MBR expression (e.g., batch size ÷ equipment working capacity) or is chosen at run time within the MBR's minimum and maximum, with a reason. Each instance gets a K-25 sub-lot ID with its own data, timers, yields and sampling (e.g., per-load sterility pulls, ST-66); the join needs every instance terminal; a failed instance leads to a sub-lot disposition (reject or QA hold) without failing its siblings. Simulation unrolls to the maximum.
- **Time constraints** are a separate list of min/max durations between any two step events, even across stages or batches (dirty hold, bulk hold, compounding start to end of filtration); per iteration inside loops and per instance inside replicate sections; void if an end lies in an untaken branch.
- **Exceptions are not arrows:** each block declares what it can raise (limit, time, status, missing verification, interface failure) and the kernel turns it into an exception record (5.2.4). Authorised roles can always hold or abort a batch. Whole-batch rework runs only as a separately approved reprocessing procedure invoked by QA; reprocessing and recovery additions are dedicated steps with a declared case (5.2.9; PR-47, IN-19).

6.3 Static verification gates

The same validator runs live in the designer and authoritatively on the server; errors block submission, warnings need a recorded justification, and each message names its rule and control (e.g., "Timer lacks validation reference (ST-37)"), in plain language first.

- **Structure:** one start and end; every step reachable and able to finish; only declared loops and replicate sections; matched splits and joins; no edge crossing a parallel, loop or replicate boundary; loops have a maximum and an exit that reads a value written inside; time constraints consistent.
- **Data:** every value written before it is read on every path; expressions parse, type-check and unit-check; no calculation cycles; branch conditions cover every case and never overlap; parallel branches never write the same value or claim the same equipment or lot unsequenced.
- **GMP content:** every BOM line consumed on every completing path; each stage starts with clearance and ends with yield/reconciliation; equipment use preceded by a status check; critical parameters have unit, precision, nested limit tiers with source and pharmacopoeia edition, and an accuracy-check method for manual entry (DI-30); performer plus checker on significant steps (PR-14); WITNESSED meaning where a critical value has no

objective evidence (5.5); timers have validation references; yield checkpoints have min and max (PR-31); reconciliation limits exist (PR-39); content minima for the market set, including India's hold time for every intermediate and approved specimens for every printed component (PR-03, IN-13, IN-14); all references effective; Established-Condition elements have a reporting category; **every risk-based reduction** (one-verifier mode S5a, no second person on clearance S24, RBE, reduced IPC frequency, "minor" classifications) carries a risk class from the controlled scale and a linked rationale (IN-02); each reprocessing or recovery step declares its case and limits (5.2.9); ALL_APPLY acceptance schemes have a sampling plan that serves every criterion.

- **Units** (L-39): hardness registered as "kg/cm²" in Indian dossiers is accepted only through a controlled legacy alias "kg/cm² (as kgf/kp)" mapped to kgf and N with a recorded conversion, raising a warning that needs justification; a genuine pressure unit on a force parameter without the alias is refused.
- **Tablets:** compression IPC interval within the resolved `tablets.compression.ipcSampleMaxInterval` (30 min, IN-35); **station-wise check at run start and at a fixed MBR interval during the run on multi-point presses** (Schedule M Part VIII 3.4; IN-35(b); test: a 16-station press with an incomplete periodic station set is flagged); tooling and sieve checks before and after use; time and temperature on every mixing, blending and drying step (IN-13); QC gate after printing (IN-37).
- **Injectables (IB-01):** contamination-control-strategy reference on every critical control point; sterility-route justification; step grades consistent with mapped rooms (IN-41, ST-14); pre-use filter test always, PUPSIT or a referenced waiver; 100 % CCI for fusion-sealed containers up to 100 mL; a sterilisation method for every product-contact item; open containers only through Grade A; mandatory sterile timer set; 100 % visual inspection; sterility pulls per load for replicate sections.
- **Also:** market-resolution run showing resolved values, governing markets and predicate branches (IN-52); translations complete.

6.4 Expression language

In plain words: calculations use a small formula language with exact decimals and units that cannot loop forever and gives identical answers in browser and server.

- Pure and deterministic; no I/O, user functions or unbounded loops; reads only declared inputs; aggregates only over recorded series.
- **Decimal** at 34 significant digits, never binary floating point (raw 94.95 against a 95.0-105.0 limit must round to 95.0 and pass).
- **Units:** UCUM plus a semantic tag (% w/w, % label claim, % LOD, % yield, count), checked at design time: adding % LOD to % yield is refused.
- **Rounding** only at limit comparison and display, half-up, to the limit's decimal places (USP) or significant figures (Ph. Eur.) per its rule set; under ALL_APPLY each criterion uses its own rounding (L-37); full precision is stored, never replaced by a rounded value.
- **Explicit errors:** missing input, division by zero or out-of-range raise an exception, never a silent zero.
- **Closed function library:** arithmetic, min/max, mean, SD, RSD, yield, potency/assay/LOD-adjusted quantity with compensating excipient (C-01..C-06), reconciliation, durations, F0 and lethality, compendial acceptance tables. Customers parameterise; they do not re-implement compendial statistics.
- **Test vectors:** boundary vectors (typical, at limits, at rounding boundaries, absent inputs) are **generated automatically**; the author confirms expected results written in plain language (L-65). Each result stores expression version, inputs with units and sources, and output. Dependents of a corrected input are classified as derived results, decision inputs or executed targets and handled per 5.2.3.

6.5 Versioning, where-used, simulation and test runs

- Every reference pinned by version and hash. Where-used answers "which CURRENT MBRs and running batches use material M, equipment class E, block B v3, SOP S rev 7 or Established Condition X"; a library or rule-table change yields an impact report and optional bulk draft versions under one change control.

- **Delta verification** (L-60): when a version differs from its verified predecessor only in parameter values, the structural evidence is reused and regenerated boundary vectors run automatically for the changed parameters; a **bulk approval package** shows a per-MBR difference, while signatures stay per MBR. RBE confidence carries over for value-only changes if the exception rules for the changed parameters are unchanged; any structural change resets it.
- Reviewers see a structured difference against the predecessor plus the full paginated rendering identical to what operators see.
- **Automated simulation** (no writes): every path with loops and replicate sections unrolled to their maximum; boundary values for every guard and limit (on, just inside, just outside, missing); calculations against golden inputs; coverage report.
- **Sandbox test runs** on the real screens against the separate sandbox database with reserved identifiers and a "TEST, NOT GMP" watermark: no real inventory, genealogy, equipment status or release, never in GMP reports; a bypass test proves the sandbox role cannot write to any GMP database (DI-05, INV-18, L-52). The same store serves training and simulation mode.
- **Generic masters:** structure version + approved parameter set; approval covers the pairing and the issued record cites both; only declared parameters vary.

6.6 Building blocks per edition

Generic blocks (recipe-core if shared), with aliases from both specs:

Canonical	Tablets	Injectables	Main controls
G-01 Issuance and header	K-01	KB-01	PR-01, PR-10, IN-18, IN-30, IN-52, DI-42
G-02 Stage frame and summary	K-02	new	PR-16, IN-15
G-03 Instruction and acknowledgement	K-03	KB-02	PR-03, DI-10
G-04 Process parameter record	K-04	KB-02	PR-08, IN-13, DI-07, DI-30
G-05 Observation capture	K-05	KB-02	PR-11, DI-28, DI-41
G-06 Automated equipment phase	K-06	KB-03	PR-14, DI-31, DI-32, DI-49
G-07 Weighing and dispensing	K-07	KB-05	PR-18 to PR-20, IN-23, IN-24
G-08 Identification and charge-in	K-08	KB-06	PR-21, DI-30
G-09 Equipment status, cleaning, logbook	K-09	KB-07	PR-22 to PR-24, PR-27, IN-25
G-10 Room/line status and clearance	K-10	KB-08	PR-25, PR-26, PR-34, IN-21, IN-26
G-11 Environmental condition check	K-11	(IB-02, IB-05)	PR-25, IN-39
G-12 Sampling	K-12	KB-09	PR-29, PR-44, IN-20, IN-32
G-13 IPC with readings and statistics	K-13	KB-09	PR-29, IN-35, DI-29
G-14 Calculation	K-14	KB-04	PR-31, DI-33
G-15 Timer, hold time, transfer	K-15	KB-10	PR-30, IN-13, IN-15
G-16 Verification / witness	service S-3	KB-11	PR-14, PR-20, PR-21, DI-30
G-17 QA hold point	K-16	KB-12	PR-45, IN-37
G-18 Exception and deviation link	service S-4	KB-13	PR-13, PR-49 to PR-51
G-19 Flow control (incl. replicate)	K-17, K-25	KB-14	DI-34, PR-13
G-20 Yield and reconciliation	K-18	KB-15	PR-31, PR-39, IN-17
G-21 Printed materials, coding, challenges, serialisation	K-19..K-21	KB-17	PR-32 to PR-42, IN-16, IN-22, IN-31
G-22 Output containers, holds, split/merge	K-22, K-25	KB-16	PR-38, PR-45, PR-57

Canonical	Tablets	Injectables	Main controls
G-23 Returns, rejects, destruction	K-23	KB-18	PR-40, PR-46, PR-48
G-24 Reprocessing / rework gate	K-24	KB-18	PR-47, IN-18, IN-19
G-25 Sign-off, review, release and market disposition	K-26	KB-19	PR-52 to PR-56, IN-04, IN-05, DI-25, DI-26, DI-39

Tablets (TB-01..24). *Dispensing and granulation:* TB-01 potency/assay/LOD-adjusted dispensing with compensating excipient; TB-02 sifting and milling with screen checks before and after; TB-03 binder/coating/sugar solutions with use-by; TB-04 wet granulation endpoint; TB-05 drying with LOD loop and filter bags; TB-06 roller compaction; TB-07 blending, lubrication, blend uniformity; TB-08 reprocessing and recovery additions (5.2.9). *Compression:* TB-09 tooling custody and inspection; TB-10 set-up and first-off; TB-11 periodic IPC scheduler with multi-channel streams, periodic station-wise sweeps and bracketing hold; TB-12 weight IPCs; TB-13 breaking force, thickness, appearance; TB-14 friability; TB-15 disintegration; TB-16 bilayer; TB-17 reject and detection devices. *Coating and finishing:* TB-18 coating and weight gain; TB-19 printing with QC approval; TB-20 inspection, sorting, AQL. *Packaging:* TB-21 blister/strip set-up with independent check; TB-22 leak test; TB-23 bottle line; TB-24 secondary/tertiary and finished-pack examination. Mainly IN-13, IN-34 to IN-39, PR-19, PR-27, PR-29, PR-33 to PR-41, PR-47, DI-29, DI-33.

Injectables (IB-01..30). *Design and readiness:* IB-01 sterile MBR validator; IB-02 line/room readiness; IB-03 personnel, gowning, aseptic roster; IB-04 EM/PM batch window; IB-05 utilities and alarms. *Preparation:* IB-06 component/RTU acceptance; IB-07 container and closure preparation; IB-08 sterilisation, depyrogenation, SIP; IB-09 sterile items, transfers, aseptic assembly; IB-10 isolator/RABS decontamination and gloves; IB-11 single-use systems. *Processing:* IB-12 compounding; IB-13 sterilising filtration and integrity; IB-14 sterile time limits; IB-15 APS status and envelope; IB-16 interventions (with unlisted entries); IB-17 filling and closure; IB-18 fill weight/volume and device IPC; IB-19 capping; IB-20 lyophilisation; IB-21 terminal sterilisation (loads as replicate sub-batches); IB-22 BFS/FFS. *Inspection and release:* IB-23 visual inspection and AQL; IB-24 CCI; IB-25 unit accountability; IB-26 event-driven sampling; IB-27 sterility and endotoxin results; IB-28 APS record; IB-29 contamination-event impact; IB-30 sterile dossier and certification gate. Every ST-01..ST-79 control and IN-40..IN-51 maps to at least one block.

Block contract (L-61): a block type is vendor code with, for each version: configuration and data schemas (versioned, with upcasters and a configuration migrator), execution handler, exception rules and their mapping to the RBE exception taxonomy, shop-floor UI, review renderer, true-copy/PDF renderer, paper-contingency rendering, simulator model for automated simulation, translations, edition tag, tests and a golden fixture. CI refuses to register a block version missing any part. Customers cannot create primitive types; a partner extension would be bespoke software (GAMP category 5), not configuration.

7. Users, segregation of duties and key screens

7.1 Persons, accounts and roles

In plain words: rules about "not the same person" are checked against real people, not login names; each person gets roles for specific sites and areas, and role, training and independence are checked at every action.

- **Persons and accounts** (L-02): a person (verified identity) owns one or more accounts; at most one active GMP-capable account per person per installation (partial unique index); a rehired contractor keeps the same person record. Every segregation rule (INV-03, INV-04, INV-35, S7, S9, S10, S14, S26, S27) is evaluated on the person. A person's administrator account may never act on that person's own GMP account or grants (extends S12). INV-11 applies across the whole installation, not per scope. In the v1.1 hub topology the hub issues the person identifier.
- **Roles:** 29 kernel roles (features A.2) in four groups: master records (author, independent checker, QA approver, regulatory-conformity signer, master-data steward); execution (batch issuer, operator, verifier, designated dispenser, shift supervisor, person responsible for a stage, head of production, shop-floor QA, labelling store, maintenance/calibration/cleaning); quality and release (QA and audit-trail reviewer, head of QC, QC analyst, India authorised person or named delegate, US QCU disposition signer, EU QP only with the switch); governance (training coordinator, system administrator, process owner, CSV lead, read-only auditor, contract-giver reviewer, service account, time-boxed vendor support). Edition skills are qualifications on operator and verifier roles, not new authority levels. Grants are scoped (entity, site, area, product, line, step type) and need a separate approver.
- **Delegation** (L-06, IN-05(b)): a delegation names delegator, delegate person, scope (site, product, market, meanings), validity window and SOP reference, and is approved by someone other than the delegate; no re-delegation; the signature stores the delegation ID, the guard checks validity at signing, and S10 applies to the delegate.

7.2 Segregation of duties

Static conflicts are refused at role assignment (administrator + GMP role; write rights on a generic account); *dynamic* conflicts are refused per record at the moment of action.

- **Involvement set** (L-04, R-23): a person is involved in a batch if, in the batch or any of its packaging orders, they signed anything other than the QA review, the audit-trail review or a release/disposition meaning (production approval counts, so the production reviewer cannot also sign the QA audit-trail review or release); entered, corrected or voided data; dispositioned an exception before EXECUTION_COMPLETE; approved a skip or N/A; signed a QA hold point; performed or verified a line clearance; or took a sample. QA in-process checks (IPC or AQL sampling) count by default (the one remaining switch, DI (f) 13). The set is computed in the review or disposition transaction from system-of-record tables.
- **Locked:** S1-S3 (author not checker, QA approves every MBR, no self-approval; editors per INV-04); S5 performer is not verifier on significant steps, dispensing, charge-in, yields and critical manual data (US rule for all markets); S6 verifier's own credentials with all components; S7 no self-approved deviation; S8 production approval before QA review; S9 production head is not quality head on a batch; **S10 involved persons barred from the batch's audit-trail review, manufacturer release and market dispositions** (removed from the switch list); S11 only the release authority releases; S12-S16 administrator separation, break-glass limits, requester is not approver, no shared write or signing service accounts, no proxy signing; **S17 only approved technical staff sign production-supervisor and QC-approval steps** (locked because every batch is India-licensed; register content configurable; Drugs Rules basis S, R-06); S18-S23 designated dispensers, authorised labelling issue, status changes by QC/QU only, QC approver is not the analyst, manual-cleaning and blister-clearance performer is not verifier; S26-S27 yield verifier and unusual-event approver independence; S29-S31 sterile approvals; **S32 export markets independent, all gated by the India manufacturer release**.
- **Configurable** (defaults in 3.5): S4, S5a (**two persons on every batch** by default, Decision 8), S24, S25, S28.

7.3 Key screens

Each screen is tagged with the phase that delivers it (P1-P7).

- **Shop floor** (P2 generic step screen, P3 tablets, P5 injectables): operator cockpit (one queue of steps, verification requests, IPCs and timers, showing why an item is not available); step screen (one action at a time; fixed header with product, batch, IN/US/EU badges, stage, room, equipment, user, timers, connection; prerequisites re-checked at commit and signing; badge-tap prompt per entry on shared devices); scan dialog; dispensing (live gauge with tolerance band, stable-reading capture, potency trace, label print, automatic two-person mode when the scale is offline); **IPC grid entry** (L-66: a reading set entered in one grid and committed once, each reading stored individually, the echo shows the whole list with units; mean, RSD and mini chart from committed values, next-check countdown, station grid); exception capture; signature dialog (lists exactly what is covered, never pre-filled); verification queue with inline post-entry changes; equipment/room board and e-logbook; late-entry and contingency screens; shift handover (P3); [T] packaging line with live reconciliation; [I] aseptic quick-log with catalogue entries and "Unlisted intervention" (5.2.11), visual-inspection station.
- **Planning and masters** (P3 unless noted): batch planning and issue (market set, size, MBR selection with CURRENT/BRIDGING explanation, reproduction check); master-data and status management (materials, lots, equipment, instruments, rooms, tooling, screens); packaging master and artwork approval with specimens (IN-14); qualification and training matrix (P1) and aseptic roster (P5); stand-in screens for deviation, sample and result entry (P3) and the qualification register (P1); delegation management (P1); rule-table site values and policy console under change control (P1); contingency-form issue and reconciliation (P4); notification centre (P3); leaver-obligations screen (P1).
- **QA review** (P4; L-62): production review first; then the exception register (all mandatory types, grouped by cause; each item Accept / Query / Escalate with keyboard "next item"); item detail (expected vs actual, limit source, audit snippet, instruction as issued, raw device data) and the full record one click away. **Grouped acceptance:** non-critical items that share one linked incident or deviation may be accepted together with one justification after the full list has been shown; each item still gets its own disposition row and the signed content lists every item, so every exception is still dispositioned individually (PR-52). **Critical items** are always individual; their detail must have been opened (tracked) and a justification is required. QA oversight: a random second review of a configurable share of reviews and review-speed metrics per reviewer. Audit-trail review panel with linked-record changes and high-water mark; "Full review" or "RBE rule set vN" badge; signature enabled only when production approval, every item and the audit-trail review are complete; amendment and reopen workflow.
- **Release cockpit** (P4): manufacturer-release checklist first, then one tab per market with the red/green G-REL items linked to evidence and a "since review" list; justified-open-investigation path only where the switch allows; EU tab with QP package delivery, "awaiting QP certification" and the structured certification intake; **post-release suspension and recall cockpit** (L-19); in-flight disposition screen for MBR changes (P2, L-24).
- **MBR designer** (P2; L-65): the primary authoring view is a **document-like outline** (stage → operation → step table) that QA and production authors recognise; the React Flow canvas is a secondary view per operation; authoring starts from library unit-procedure templates, with wizards for IPC plans and dispensing BOMs; edition-only palette with pinned versions; property panel; formula editor with generated boundary vectors and plain-language expected results; operator preview per language; validation panel in plain language citing control IDs; market panel (resolved values, governing markets, per-market authorisation); where-used; difference view; test-run launcher with coverage; approval view as a paginated document identical to what operators see plus the structured difference (Prepared → Checked → Approved).
- **Dashboards** (P6: right-first-time, exception Pareto, review cycle time, deviation ageing, hold-time risk; [T] IPC SPC, press OEE; [I] EM trends, interventions vs APS envelope, APS due dates, inspection rejects), each showing data age and drilling to the record, never used for GMP decisions. Also: genealogy explorer (P6), audit-trail viewer (P1), administration console (P1), read-only inspection workspace (P4), PQR/APR extract screen (P4).

7.4 Accessibility, languages, shared devices and gloved use

- **Accessibility:** WCAG 2.2 AA as target; shop-floor targets at least 44 x 44 CSS px (primary actions about 15-20 mm for double gloves); state shown by colour + icon + word; contrast at least 4.5:1; keyboard and screen-reader support on office screens; warning before inactivity lock, extended with one tap on personal devices and with a badge tap on shared devices; automated axe checks in CI.

- **Shared devices** (L-01): per-entry attribution as in 5.5; identity banner always visible; fast user switching by badge; an unsaved entry never carries over to the next user (4.2).
- **Languages:** English + Hindi in v1; regional languages per customer (candidates Gujarati, Marathi, Telugu, Tamil, Kannada, Bengali) with full Indic shaping. MBR translations are reviewed bilingually and approved with the MBR version; machine translation only as draft; the record stores the language shown; language never changes a limit, unit or meaning. GMP values in Western digits with SI symbols; lakh/crore grouping only as display option; parsed numbers echoed with unit before commit; dates like 27-SEP-2026. **Free text** (R-34): for any batch with US or EU in the market set, free-text GMP fields (deviation descriptions, comments, justifications) are entered in English or carry a verified English translation (translator, verifier, both texts kept) before production approval; an untranslated entry is an RBE exception type (policy, default on; 3.5).
- **Gloves and cleanrooms:** scanner first, keyboard last; numeric keypads with fixed units; no drag, multi-finger or hover-only controls; large fonts readable through visors; [I] paperless Grade A/B on registered GRADE_A_B devices with quick-log buttons. Gowned usability tests, including the signature budget per golden batch (5.5), happen with pilot customers.

---

8. Helpful extras (ranked) and what is deferred

Ranked by value against effort (features spec C). None turns a block into a warning or takes a GMP decision. The **PQR/APR extract is no longer an extra**: it is core scope in Phase 4 because IN-01, PR-58 and ST-73 require structured, reproducible extracts in v1 (R-22).

Rank	Extra	Plan
1	Scan-everything with GS1 and Indian API QR parsing (manual identity entry is an exception needing verification)	v1
2	Right-first-time and exception analytics	v1
3	Operator cockpit with server-side timers and escalation	v1
4	[I] Consolidated sterile timeline for review	v1
5	Genealogy visualisation with mock-recall simulator	v1
6	Training/simulation mode (separate sandbox database)	v1
7	Hindi UI (regional languages per customer)	v1
9	Change-impact explorer	v1
11	Paper-fallback kit generator (pre-generated packs; delivers mandatory DI-43)	v1
15	Certified evidence export supporting a BSA s.63 certificate (format to confirm)	v1
13	Class 3 DSC / eSign for key signatures	interface v1, off by default; provider after legal opinion (Decision 3)
8, 10, 14	Batch comparison overlay; [I] EM mobile sampling app; camera-based code verification	v1.1
17-19	CPV/SPC dashboards (the PQR extract itself is v1); rich-media instructions; BatchML/B2MML export	v1.1
16, 22	Foot pedal/voice for non-critical confirmations (never signatures, values or identity); [T] MACO/HBEL calculator	later
12	Offline terminal mode under DI-28	deferred (Decision 12)
20, 21	AI-assisted MBR digitisation and review hints (advisory, logged, never the review)	after EU Annex 22 is final
23-25	Predictive ETA; graph analytics copy; AGV transport	deferred

Never offered: a "skip review for testing" switch, free scripting, or anything that turns an interlock into a warning.

9. Quality and validation strategy

In plain words: correctness is shown by thousands of automatic tests that attack the rules from every side, and the validation documents are generated from those same tests, so they stay current.

9.1 Test layers

Layer	What it proves
Unit (Vitest)	Each guard, calculation, limit evaluation, rounding rule, comparison method, block handler and adapter
Property-based (fast-check)	The evaluator matches an exact-rational reference; unit conversions round-trip and dimension errors are refused; USP and Ph. Eur. rounding; hashing ignores key order and any signed-field change alters the hash; limit tiers nest; resolution never looser than India for every atomic key; every random block-structured flow (including replicate sections) compiles to a sound graph and every broken one is rejected with the right rule
Model-based (XState paths, fast-check commands)	Every transition and guard-false branch replayed against the real API and database; random command sequences (issue, record, correct, void, invalidate, sign, verify, hand over, hold, abort, deviation, release, suspend, add or remove a market) with all invariant queries returning zero rows after every command; race interleavings (two operators on one lot, correcting while someone signs, a deviation linked while a release commits)
Database bypass	Split into **prevented** (raw SQL as the application role cannot update a data point, delete an audit row, overlap occupancy, update a status column, insert a signature or an authentication event, write from the sandbox role, insert a shared-device data point without an authentication event) and **detected** (a superuser rewrite is caught by chain verification against anchors; a transplanted or forged signature by seal verification) (L-55)
Golden batches	End-to-end batches with exact expected outputs (states, exception report, quantities, yields, genealogy, record hash, normalised true copy), each with happy and exception-rich variants
Browser end-to-end (Playwright)	Real screens as each role, producing step-by-step evidence (expected, actual, screenshot, time) as PDF test records
Mutation (StrykerJS)	At least 90 % score for guards, signature invalidation, limits, rounding, flow validation and release gating; 80 % elsewhere; survivors in high-risk code need a test or written justification
Non-functional	Load (50,000-entry batch review under 5 s; p95 command under 300 ms on site LAN; release under concurrent EM and data-point ingestion without repeated retries), restore drill with chain verification and restore event, timers with job workers stopped, accessibility, translation completeness, dependency and licence scans
Usability	Designer build of the GB-T-01 MBR by 3 QA and production authors (Phase 2 exit); signature budget and task times per golden batch; gowned tests with pilot customers

Golden batches. Tablets GB-T-01 from the user's sample BMR: high-shear wet granulation in a **split granulation** (replicate section), fluid-bed drying with LOD loop, film coating, 2,00,000 tablets; salt-to-base potency with compensating excipient; optional extra-IPA branch; **reprocessing addition** of milled rejects limited to 3 % of batch and 3 months age (case 2 of 5.2.9); left/right IPC with a failing check that holds overlapping containers; "approved for coating" and "release for packing" gates; inspection rejects and witnessed destruction. The sample's data problems become tests: hardness in kg/cm² accepted only through the declared legacy alias with a warning, a genuine pressure unit refused; trihydrate label on an anhydrous molar mass flagged; an addition without its declared case and approval refused; its (A + B)/T yield shown beside the good-output yield. Also direct compression, roller compaction, bilayer, blister with printed-material reconciliation and two packaging orders from one bulk (full clearance between them), bottle; a correction of a failing IPC without evidence refused; US added after an India-only vendor-history excipient lot was consumed (INELIGIBLE, the identity test should have preceded use). Injectables: isolator vial fill with EM and interventions, lyophilised vial with two dryer loads, terminally sterilised ampoules in three autoclave loads, APS; negative variants: EM action result sampled on day 1 and reported on day 5 holding the batches of days 2-4, failed post-use filter test, unlisted intervention, APS expiring during a fill, one positive in 12,000 on a line gaining its first EU product. Cross-cutting: a site running both editions sharing site services; a restore to an earlier point.

9.2 Validation approach and deliverables

- **GAMP 5 category 4** platform, validated once per vendor release, plus **verification of every MBR**, library element, workflow, report, calculation, interface mapping and RBE rule set before it becomes effective (DI-04: qualifying the platform does not prove a given electronic master is right). No script escape hatch, so configuration never becomes bespoke code.
- **CSA** (FDA, 2025) only as informative rationale for scaling rigour; the binding basis is 21 CFR 211.68 and Part 11, EU Annex 11 and Schedule M Part I section 20. High-risk functions (limits, calculations, signature binding, attribution, status and release interlocks, record generation, audit trail, interfaces, RBE rules) get scripted tests with evidence plus exploratory sessions; low-risk ones (layout, dashboards, notifications) unscripted or scenario tests with recorded outcomes.
- **Review by exception** (L-62, DI-26): the rule set is validated with seeded-exception tests with zero false negatives **on sandbox copies only**, never by planting events in GMP records; an MBR earns RBE by running full review and RBE in parallel on real batches and comparing the findings against a confidence criterion; periodic challenge by comparing sampled full audit trails with the filtered view.
- **Generated per edition in CI:** URS from the register; risk assessment per requirement (ICH Q9(R1)); functional and design specifications from the code (state diagrams, guard tables, block catalog, permission matrix, constraint, privilege and trigger inventory, audit event catalogue); traceability matrix; IQ script (versions, image digests vs SBOM, triggers, constraints and privileges enabled, locked settings); OQ = regression and golden suites on the customer environment; PQ templates on the customer's MBRs; validation summary; release notes with requirement-level impact; SBOM; known issues.
- **Kept current:** each pull request names its control IDs and risk class; CI regenerates everything, fails if a high-risk requirement within its target phase lacks a passing scripted test, and emits a validation delta per release for customers' change control.

---

10. Modifiability

10.1 Practices

In plain words: rules live in data files, processes are built from plug-in blocks, life cycles are explicit maps and decisions are written down, so a future developer can change one thing without breaking another.

- **Rules as data** in `compliance/` (controls, kernel and edition rule tables as rows, floor tables, pharmacopoeia sets, policies, signature meanings, exception taxonomy, reason codes), schema-checked in CI.
- **Plug-in blocks** following the block contract (6.6), registered by code and version.
- **State charts:** one definition per life cycle; the transition table, diagrams and tests are generated from it (a test fails if they differ).
- **ADRs** (`docs/adr/`): 001 two editions, separate edition databases, shared site services; 002 PostgreSQL only, no graph database; 003 TypeScript end to end; 004 block-structured flows with replicate sections; 005 in-house decimal/UCUM expression language; 006 server-authoritative execution, no device signing; 007 hash-chained audit trail with checkpoints, anchors and restore events; 008 market-profile engine with atomic keys and stored resolutions; 009 control register as data; 010 versioned execution handlers, retained renderers; 011 recipe-core sharing (after Decision 1); 012 privilege separation and SECURITY DEFINER transitions; 013 manufacturer release plus market dispositions; 014 per-market MBR effectivity; 015 published, exactly pinned shared packages.
- **REPO-GUIDE.md:** where things live, how to run locally, how to add a block, limit, market or rule, definition of done.
- **Coding standards:** TypeScript strict; JavaScript `number` banned for GMP values; `Date.now()` banned in domain code; no cross-edition imports; reviewed, linted SQL migrations; Conventional Commits; PR template asking for control IDs, risk class and validation impact; CODEOWNERS second review for `packages/kernel`, `packages/site-services`, `compliance/` and migrations.
- **CI on every change:** type-check, lint and boundaries; unit, property, model-based and bypass tests; golden batches; browser smoke tests; incremental mutation testing (full weekly); register and table schemas; behaviour-key India review check; trace-matrix completeness within target phases; machine/transition-table consistency; accessibility; dependency, licence and secret scans; SBOM; each edition built against its pinned packages plus a HEAD canary.

10.2 How a future developer...

- **Adds a block:** copy the block template; supply every part of the block contract; add validator rules if needed; list the controls in their register files; register a new block-type version. Existing MBRs are untouched until an author uses it.
- **Changes a limit:** a product limit is not a code change: the author edits it in a new MBR version (delta verification and approval). A regulatory value is a new row in the rule or floor table (value, floor type, source, verification status, effective-from); the new table version produces tracked change tasks for affected MBRs and site values; a common limit across hundreds of MBRs goes through one bulk approval package with per-MBR signatures (6.5).
- **Adds a market** (e.g., WHO-GMP, UK) (L-58). *Data:* the market code, rule rows with sources, label and serialisation rules per pack destination, PIC/S mapping. *Code plus a validation delta:* the market's release authority and disposition states (like AWAITING_QP_CERTIFICATION), its G-REL items, labelling and serialisation adapters, and any new comparison method; the resolution test matrix then covers the new market in every combination.
- **Updates a rule after a regulation changes:** the section 3.6 workflow.

11. Phased build plan

In plain words: foundation and security first, then one complete product, then the second; every phase ends with something you can watch working and passing tests, and each simulator or stand-in arrives in the phase that first needs it.

Each phase ends with a tagged release, a demo script, green CI, the trace matrix updated for controls whose target phase it is, and release notes listing controls still pending the expert. Effort shares are rough; Injectables is weighted above Tablets for its 30 blocks and 79 sterile controls (L-50).

- **Phase 0, repo and CI (~3 %).** Monorepo, strict TypeScript, boundary lint, local PostgreSQL scripts, CI with every check stubbed; private registry with exact pinning (4.3); register import with target phases, decision import, trace-matrix generator; ADR drafts, REPO-GUIDE, PR template. *Exit:* register schema valid; a boundary violation fails CI; an edition skeleton builds against a published kernel.
- **Phase 1, kernel, site services and security (~16 %).** Persons, accounts, identity proofing, local accounts (argon2id), optional OIDC with step-up rules, badge + PIN, MFA for privileged access, sessions, device registry and identity, shared-device attribution, roles, delegations, person-based segregation, leaver processing; **qualification register stand-in**; audit capture, chains, checkpoints, anchors, restore events, verifier and viewer; authentication service, SECURITY DEFINER signing and transitions, seal verifier; two-timestamp time model; idempotency; outbox; state-machine engine; market-profile engine with **stub lines, rooms, equipment and lots**; policy versioning; retention with dependencies; true-copy bundle and verifier CLI; DI-01 register, incidents, DI investigations; **threat model, ASVS L2 design review, HA and restore design with a drill** (4.2, 4.7). *Exit:* INV-03, 06-11, 16, 22, 26-29, 40, 47 with bypass tests; DI-01, DI-12 to DI-24, DI-27, DI-35 to DI-38, DI-45, DI-47, DI-50 and IN-52 (stub scopes) tests pass; kernel mutation targets met; design review signed off.
- **Phase 2, recipe engine and designer (~14 %).** Expression language with generated boundary vectors; flow model with replicate sections, validator (including the IN-02 rationale gate), engine, simulator; limit tiers, rounding, ALL_APPLY; **engine-level generic blocks** (G-01 to G-05, G-07, G-08, G-11 to G-20, G-22 to G-24; G-06 and G-21 finish in Phase 3, G-25 in Phase 4); MBR life cycle with per-market authorisation, bridging and in-flight disposition; libraries, templates, generic masters, where-used, difference view, delta verification, bulk approval; sandbox database; **generic shop-floor step screen**; document-first designer; **site-gateway core and scale simulator**. *Exit:* INV-01, 02, 04, 17-20, 25, 34 (generic part); DI-04, DI-05, DI-33, DI-34, PR-01 to PR-08, IN-02 tests pass; designer usability test with 3 authors passed.
- **Phase 3, Tablets edition (~17 %).** TB-01..TB-24; G-06 with **press and coater simulators**; G-21 for blister packaging with the **printer simulator** and a serialisation stub; equipment kinds, campaigns, clearance types; lots with release routes and reservations; containers; reprocessing and recovery; execution with exception tiers, correction/void/invalidation paths and hand-over; **ERP stub, LIMS-lite and deviation-lite**; shop-floor, planning and master-data screens. *Demo:* GB-T-01 (9.1). *Exit:* INV-12 to 15, 24, 31-33, 41; IN-13, IN-15, IN-17, IN-21 to IN-26, IN-34 to IN-39 and the tablet PR controls tested; GB-T-01 matches expected outputs exactly.
- **Phase 4, review, release and India specifics (~13 %).** Production and QA review (7.3); audit-trail review with linked records and high-water mark; G-REL with evidence epochs; manufacturer release and market dispositions, EU QP package and certification intake; ERP shipping interlock; post-release suspension, recall cockpit, rejection and disposal; market-add evaluation and removal; full deviation stand-in or QMS adapter; amendments and locking; India fields (licences and loan licence, allocation register, India-domestic label, Schedule H2 QR payload, export permission); retention; PDF/A-3 true copy; evidence export; contingency packs and form reconciliation; **PQR/APR extract** reconciled to the batch, deviation and CAPA registers, export-only SKUs included, with AP sign-off; free-text language policy. *Demo:* as in the summary table. *Exit:* INV-05, 21, 30, 35, 38, 39, 42, 43, 46, 48; PR-09, PR-44, PR-52 to PR-59, DI-25, DI-26, DI-39 to DI-47, IN-01, IN-03, IN-05, IN-08, IN-11, IN-14, IN-16, IN-18, IN-29 to IN-32 tests pass.
- **Phase 5, Injectables edition (~22 %).** IB-01..IB-30; clean-area hierarchy, EM linkage, trends and completeness (5.2.10); **EMS/BMS, autoclave, lyophiliser, particle-counter and access-control simulators**; sterilisers, filter trains, technology profiles, intervention catalogues with the unlisted path; APS status, envelope and record machines (5.2.11); line-tightening gate (3.3.4); sterile timers, dossier and timeline; sterility-assurance investigations locked out of the PR-53 path. *Demo:* as in the summary table, plus an unlisted intervention needing

real-time QA authorisation. *Exit:* INV-23, 36, 37, 44, 45; all ST-01..ST-79 and IN-40..IN-51 tests pass, with named tests for ST-19/ST-20 (continuous Grade A counting; ISO 8 vs Grade D), ST-21 (sampling-to-reporting linkage), ST-29/30/33 (granularity, manual per operator, EU failure), ST-31 (in-fill envelope), ST-32 (unit accountability; FDA incubation labelled FDA), ST-52 (redundant and serial trains), ST-58 (manual-open lyophiliser re-sterilised before every load), ST-64, ST-73; a bypass test of the justification path on an open Grade A investigation.

- **Phase 6, integrations and extras (~8 %).** Full adapters with contract tests (ERP/WMS, QMS, LMS, CMMS, EMS/BMS, OPC UA, scales, printers, serialisation L3/L4) and fault injection including stale mirrors; dashboards; genealogy explorer and mock recall (federated with as-of points); inter-site transfer packages; Hindi + one regional language; training mode; change-impact explorer; controlled-substance mode if Decision 14 = A. *Exit:* DI-20, DI-31, DI-32, DI-49, PR-57, IN-29 tests pass; no silent loss under fault injection; stale mirrors fail closed.
- **Phase 7, validation pack and hardening (~7 %).** Generated URS, risk, specifications, trace matrix, IQ/OQ scripts, PQ templates, summary report; load and soak tests including release under concurrent ingestion; restore drill and record; penetration testing and hardening verification; accessibility audit; user and admin guides; v1.0-rc. *Exit:* every FOLLOW/MODIFY control VERIFIED; mutation targets met; no open high-severity findings.

11.1 What can and cannot be demonstrated here

- **Can:** everything above on this machine (Node 22, pnpm 10, PostgreSQL 16, Playwright; 4 vCPU, 15 GB), both editions side by side with a shared site-services database (e.g., ports 3001/5173 and 3002/5174), simulators for all equipment and systems, local accounts, simulated badges and device keys, a local signing key and an append-only directory standing in for write-once storage, labelled demo-only.
- **Cannot here** (needs a pilot customer or later step): real scales, presses, sterilisers, access-control systems and OPC UA servers; an enterprise identity provider, HSM or cloud KMS; real WORM storage; real badge readers and DSC tokens or an eSign provider; Docker deployment (the daemon is not running here, so the demo uses PostgreSQL directly); gowned usability tests; penetration testing by an external party; IQ/OQ on customer infrastructure; the Indian legal opinions (IT Act, DPDP, CERT-In, NDPS, export permissions); pharmacopoeia verification against current texts; a clean Schedule M copy; Annex 16 text.
- **Scale:** for a conventional team this is in the order of a year or more of work for several experienced engineers before a customer could validate v1; the plan front-loads the parts that carry regulatory and security risk.

---

12. Risks and mitigations

#	Risk	Mitigation
1	Regulatory interpretation errors (secondary, unverified or draft items; bridging MBR validity, concurrent PPQ release for India, QP package content)	Expert review of all controls and `proposed_modify` items; verification status carried into code and tables; watch list; nothing unverified presented as a floor
2	Missing or scrambled Schedule M pages (IN F-1 to F-12)	Obtain a clean official copy; affected controls tagged; values marked P~ or U; the US/EU rule applies meanwhile; re-check before validation documents cite them
3	Unverified pharmacopoeia values (weight bands, disintegration, friability, USP 905 constants, IP 2026)	Shipped as UNVERIFIED master data, unusable for GMP batches until QA confirms each value; source-correction path; tests check the mechanism, not the numbers
4	Legal questions in India: password e-signatures under IT Act s.5; export permissions and Rule 94; NDPS record duties; DPDP and CERT-In obligations	Part 11/Annex 11-grade signatures everywhere; DSC/eSign plug-in off until a licensed provider is configured (Decision 3); counsel questions D.5, G-9, G-10 and the 3.6 watch-list items answered before go-live
5	Scope size	Strict v1 scope (Decision 7); shared recipe-core (Decision 1); vertical slices with demos; extras deferred
6	Performance and volume at large sites	Benchmarks done for genealogy and audit writes; partitioning by batch month and closure period; row-hash audit references; archive packages; evidence epochs instead of long serialisable reads; Phase 7 load tests against SLOs
7	Subtle errors in interacting state machines (release cascade, suspension, market-add, corrections)	Model-based and property-based tests, database invariants, bypass and mutation tests; optional small formal model of sign-correct-invalidate-release-suspend
8	Integration variety and stale external data	Adapter contracts and contract tests; system-of-record table with freshness bounds that fail closed; fault-injecting simulators; flagged, verified manual fallback
9	Shop-floor usability with gowned users, including signature and badge-tap load	Glove-first design system; signature budget per golden batch; group signing within a signature point; formative tests with pilot customers before go-live
10	EU drafts change after build	Stricter drafts already followed; affected controls named (3.6); versioned tables and controls
11	Open-source supply chain and security	Pinned lockfile, few dependencies, SBOM, vulnerability and licence scans, signed artefacts; threat model and design review in Phase 1
12	Cost of keeping old execution handlers and renderers	Renderers and frozen canonical renderings kept for retention; execution handlers retired when no open batch uses them; upgrade policy with semantic-change report
13	Complexity of site services and reservations for sites running both editions	Decision 13; reservation protocol covered by model-based tests; a single-edition site runs the same code path

13. Decisions needed from you

Phases 0-2 can start on the recommendations; items that are policy switches can change later without rework.

1. **Share recipe-core between editions?** (A) share kernel, site services and recipe-core, each edition pinning exact published versions; (B) share only kernel and site services; (C) share now, fork later if needed. **Recommend A:** the flow engine, expression language and limits hold the most error-prone logic; one copy is tested and validated once, and each edition still takes changes only through its own change control. Also confirm the **kernel boundary**: persons, identity and access, e-signatures, audit trail and time plus the compliance plumbing (hashing/sealing, control register, market-profile engine, retention, true copies), none of which holds process logic.
2. **Tiered FDA/India media-fill acceptance on lines with no EU-market or PIC/S-destination product?** (A) offer the tiers there; the first EU product on such a line then triggers the evidence re-check of 3.3.4; (B) "any contaminated unit fails" on every line, tiers out of v1. **Recommend B:** stricter, simpler to validate, and most exporters serve both markets.
3. **DSC/eSign for Indian signatures?** (A) password e-signatures only; (B) optional DSC/eSign per meaning, **off until a licensed certifying authority or eSign provider is configured and its chain validates to the CCA root in OQ**, then recommended for MBR and packaging-master approval and AP release; (C) DSC/eSign for all signatures. **Recommend B**, with a test certificate authority in v1 and a banner on India-licensed tenants while counsel questions D.5 Q1-Q3 are open; switching on earlier would either block signing or produce signatures without standing under IT Act s.5.
4. **Pressure-differential defaults.** (A) 10 Pa between grades with the site's qualified value governing: this satisfies Annex 1's minimum guidance value and the lower end of FDA's "at least 10-15 Pa" example; 12.5 Pa from aseptic rooms to unclassified areas is stored as an FDA guidance example adopted as our default in all profiles (loosening needs QA approval and a CCS reference); (B) ship 12.5 or 15 Pa between grades for US exporters. **Recommend A.**
5. **Continuous particle-count evaluation.** Annex 1 9.17 compares each individual sample result with alert and action limits; Table 5 note 2 lets isolated ≥5 µm counts be treated as possible false counts while consecutive or regular low counts are investigated. One count in a one-minute 28.3 L sample is about 35/m3, above both EU 29/m3 and India 20/m3, so the question is not India-specific. (A) every single count is an excursion; (B) annotate isolated counts and escalate consecutive or recurring ones per note 2, with QA-approved window parameters; (C) a rolling cubic-metre window. **Recommend B**; the ≥0.5 µm limits and India's 5 µm values still apply per sample.
6. **IP weight bands for export-only tablet batches.** (A) IP bands on every India-made batch (tighter wins); (B) only the export market's pharmacopoeia. **Recommend A** until Indian QA or regulators confirm otherwise (G-8).
7. **v1 scope.** Tablets without capsules or continuous manufacturing; Injectables V-1, V-2, V-4, V-5, V-9, V-10 fully; V-3, V-6, V-7 blocks with golden batches in v1.1; V-8 in v1.1; no parametric release, powder filling or biologics. **Recommend accept**, or name which of capsules, prefilled syringes or BFS your customers need first.
8. **One human verifier when validated automated equipment dispensed or charged (211.68(c))?** (A) two persons for dispensing and charge-in on **every batch in every profile** by default; one-verifier mode per 211.68(c) step type only after the customer's documented regulatory position; (B) one-verifier mode by default. **Recommend A.** Zero-human verification is never allowed, and a batch run in one-verifier mode on another step type can never add US.
9. **Batch sizes and generic masters.** (A) discrete approved sizes only, with "structure + parameter set" approved as the master; (B) validated size ranges with linear scaling. **Recommend A** (B is open question PR F-11).
10. **Review by exception.** (A) build it in v1, full review by default until an MBR meets a confidence criterion earned by running full review and RBE in parallel on real batches (seeded challenges only on sandbox copies; confidence carries over for value-only changes); (B) full review only. **Recommend A.**
11. **Build or integrate LIMS, QMS, LMS, EMS?** (A) integrate only; (B) integrate plus minimal stand-ins (qualification register in Phase 1, sample/result entry and deviation record in Phase 3, full deviation workflow in Phase 4) so a site can run without them; EMS always imported. **Recommend B.**

12. **Topology, deployment and offline.** (A) one installation per site (on premise, or single-tenant private cloud for a site that accepts WAN dependency), pre-staged paper fallback, no device signing, enterprise hub in v1.1; (B) add an offline terminal mode under DI-28 in v1; (C) one company-wide installation for many sites. **Recommend A**; revisit B with poorly connected pilot sites and C with the hub.
13. **Sites running both editions.** (A) one site-services database for persons, accounts, qualifications, lots and their status, batch numbers and devices, with reservations so each edition's transactions stay local; (B) external ERP/LIMS/LMS own these data and both editions mirror them; (C) both editions in one database per site. **Recommend A:** one directory and one lot status per site while keeping separate edition databases; B needs systems many sites lack; C ties both editions to one kernel version.
14. **Controlled substances (India NDPS; importing-market controls).** (A) a product-master flag in v1 inserting dual-custody dispensing, zero-unexplained-difference reconciliation of the controlled API at each yield checkpoint, witnessed destruction of rejects and an export of quantities for the statutory registers, with a draft control IN-53 at status U until the NDPS Act and Rules are verified; (B) exclude such products from v1. **Recommend A** (Phase 6) if your customers make such products, otherwise B.

Defaults unless you say otherwise: the India AP manufacturer release gates every market; a QA person who did in-process checks counts as involved; all signature components at every signing; MBR checker and QA approver are different people; two persons for dispensing and charge-in on every batch; no grace on the 6-month APS interval; the first batch after a change waits for post-change evaluation; in-system EU QP certification off; EU-destined lots ship only to the named importer in quarantine; retention stays open until the EU certification date is received; free text on US/EU batches in English or with a verified translation; release with an open investigation blocked.